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1 **Medical image segmentation in a multiple**
2 **labelers context: Application to the study of**
3 **histopathology**

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Medical image segmentation in a multiple labelers context: Application to the study of histopathology

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27 **Segmentación de imágenes médicas en un**
28 **contexto de múltiples anotadores:**
29 **Aplicación al estudio de histopatologías**

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ABSTRACT

60 Breast cancer remains one of the deadliest diseases affecting women worldwide,
61 making accurate diagnosis and treatment planning crucial for improving patient
62 outcomes. This thesis presents a novel approach to Image Semantic Segmentation
63 (ISS) in medical imaging, with a particular focus on histopathology images of breast
64 cancer tissue. The work addresses the critical challenge of handling multiple
65 annotators' labels in crowdsourcing scenarios, where annotator performance
66 varies across the input space and image quality. The proposed solution introduces
67 a novel loss function that implicitly infers optimal segmentation and assesses
68 labeler reliability without requiring explicit performance inputs. This is achieved
69 through an intermediate reliability map that enables the model to prioritize
70 information from reliable labelers while disregarding noisy labels.

71 The methodology diverges from conventional CNN-based segmentation models by
72 eliminating the need for direct supervision of labeler performance, thereby
73 enhancing its generalizability and adaptability across diverse datasets and labeler
74 cohorts. The approach combines the strengths of U-shaped deep learning
75 architectures with a sophisticated loss function that models annotator reliability
76 across the input space. The model's performance is evaluated on both synthetic
77 datasets and real-world histopathology images, demonstrating superior results
78 compared to state-of-the-art approaches in terms of segmentation accuracy and
79 uncertainty quantification.

80 The research makes significant contributions to the field by providing a robust
81 solution for handling inconsistent annotations in medical image segmentation,
82 particularly valuable in histopathology where expert annotations are expensive
83 and time-consuming to obtain. The proposed approach shows promising results in
84 reducing annotation costs while maintaining high segmentation accuracy, making
85 it particularly relevant for clinical applications where precise segmentation of
86 tissue structures is crucial for diagnosis and treatment planning.

87 **Keywords:** Crowdsourcing, Multiple Annotators, Histopathology, Breast Cancer,
88 Semantic Image Segmentation, Gaussian Processes, Deep Learning, Medical Image
89 Analysis

91 El cáncer de mama sigue siendo una de las enfermedades más mortales en la
92 población femenina en todo el mundo, lo que hace que el diagnóstico preciso y la
93 planificación del tratamiento sean cruciales para mejorar las condiciones de los
94 pacientes. Esta tesis presenta un enfoque novedoso para la Segmentación
95 Semántica de Imágenes (ISS) en imágenes médicas, con un enfoque particular en
96 imágenes histopatológicas de tejido de cáncer de mama. El trabajo aborda el
97 desafío crítico de manejar las etiquetas de múltiples anotadores en escenarios de
98 crowdsourcing, donde el rendimiento de los anotadores varía a través del espacio
99 de entrada y la calidad de la imagen. La solución propuesta introduce una función
100 de pérdida novedosa que infiere implícitamente la segmentación óptima y evalúa
101 la confiabilidad de los etiquetadores sin requerir entradas explícitas sobre su
102 rendimiento. Esto se logra a través de un mapa de confiabilidad intermedio que
103 permite al modelo priorizar la información de etiquetadores confiables mientras
104 descarta etiquetas ruidosas.

105 La metodología se aparta de los modelos de segmentación basados en CNN
106 convencionales al eliminar la necesidad de supervisión directa del rendimiento de
107 los etiquetadores, mejorando así su generalización y adaptabilidad a través de
108 diversos conjuntos de datos y cohortes de etiquetadores. El enfoque combina las
109 fortalezas de las arquitecturas de aprendizaje profundo en forma de U con una
110 sofisticada función de pérdida que modela la confiabilidad de los anotadores a

111 través del espacio de entrada. El rendimiento del modelo se evalúa tanto en
112 conjuntos de datos sintéticos como en imágenes histopatológicas del mundo real,
113 demostrando resultados superiores en comparación con los enfoques más
114 avanzados en términos de precisión de segmentación y cuantificación de
115 incertidumbre.

116 La investigación hace contribuciones significativas al campo al proporcionar una
117 solución robusta para manejar anotaciones inconsistentes en la segmentación de
118 imágenes médicas, particularmente valiosa en histopatología donde las
119 anotaciones de expertos son costosas y requieren mucho tiempo para obtenerlas.
120 El enfoque propuesto muestra resultados prometedores en la reducción de costos
121 de anotación mientras mantiene una alta precisión de segmentación, lo que lo hace
122 particularmente relevante para aplicaciones clínicas donde la segmentación precisa
123 de estructuras tisulares es crucial para el diagnóstico y la planificación del
124 tratamiento.

125 **Palabras clave:** Crowdsourcing, Múltiples Anotadores, Histopatología, Cáncer de
126 Mama, Segmentación Semántica de Imágenes, Procesos Gaussianos, Aprendizaje
127 Profundo, Análisis de Imágenes Médicas

129	Acknowledgements	vii
130	Abstract	ix
131	Resumen	xi
132	Contents	xvi
133	List of figures	xx
134	List of tables	xxi
135	List of algorithms	xxiii
136	Abbreviations	xxv
137	1 Introduction	1
138	1.1 Motivation	1
139	1.2 Problem Statement	6
140	1.2.1 Variability in Expertise Levels	8
141	1.2.2 Technical Constraints and Image Quality	9
142	1.2.3 Research Question	9
143	1.3 Literature Review	11
144	1.3.1 Facing Annotation Variability in Medical images	12

145	1.3.2	Facing Noisy Annotations and Low-Quality Data	19
146	1.4	Aims	21
147	1.4.1	General Aim	21
148	1.4.2	Specific Aims	23
149	1.5	Outline and Contributions	24
150	2	Conceptual preliminaries	27
151	2.1	Modern Concept of Digital Image	27
152	2.1.1	Types of Digital Images	27
153	2.1.2	Mathematical Representations	28
154	2.2	Digital Histopathological Images	31
155	2.2.1	Whole Slide Imaging (WSI)	31
156	2.2.2	Regions of Interest (ROI)	33
157	2.2.3	Staining Techniques	33
158	2.3	Deep Learning Fundamentals	34
159	2.3.1	Learning Paradigms	35
160	2.3.2	Architecture and Training	36
161	2.3.3	Challenges and Solutions	36
162	2.3.4	Deep Learning Frameworks	37
163	2.4	Datasets and Data Sources	37
164	2.4.1	Oxford-IIIT Pet Dataset	39
165	2.4.2	Real histopathology datasets	39
166	3	Chained Gaussian Processes	43
167	3.1	Background and Related Methods	43
168	3.1.1	Kernel Alignment-Based Annotator Relevance Analysis (KAAR)	43
169	3.1.2	Localized Kernel Alignment-Based Annotator Relevance	
170		Analysis (LKAAR)	44
171	3.1.3	Regularized Chained Deep Neural Network (RCDNN)	45
172	3.1.4	Chained Gaussian Processes Model	46
173	3.2	Tooling and Implementation	47
174	3.2.1	GPflow Overview	48

175	3.2.2	GPflux Framework	49
176	3.3	Experimental Setup	50
177	3.3.1	Classification	50
178	3.3.2	Regression	52
179	3.4	Results	53
180	3.5	Conclusion	57
181	3.5.1	Comparative Analysis	57
182	3.5.2	Implementation Considerations	58
183	3.5.3	Final Remarks	60
184	4	Truncated Generalized Cross Entropy for segmentation	61
185	4.1	Loss Functions for Multiple Annotators	61
186	4.1.1	Generalized Cross Entropy	62
187	4.1.2	Extension to Multiple Annotators	63
188	4.1.3	Reliability Maps and Truncated GCE	64
189	5	Chained deep learning for image segmentation	67
190	5.1	Introduction	67
191	5.2	Using U-Net as a Building Block	67
192	5.2.1	Backbone Architecture	68
193	5.2.2	U-Net Architecture	68
194	5.2.3	Scalar Reliability Estimation	69
195	5.2.4	Feature-based Reliability Estimation	70
196	5.2.5	Pixel-wise Reliability Estimation	71
197	5.2.6	Integration with $TGCE_{SS}$ Loss	71
198	5.2.7	Training Process	73
199	5.3	Experimental Setup	75
200	5.3.1	Noisy Annotations Generation	75
201	5.3.2	Missing Labels Emulation	77
202	5.4	Results	79
203	5.4.1	Oxford-IIIT Pet Dataset	79

204	6 Final Remarks	83
205	6.1 Conclusions	83
206	6.2 Future work	84
207	6.3 Academic Discussion	87
208	Bibliography	89

LIST OF FIGURES

210	1-1	Estimation of the popularity of tasks and medical image types based	
211		on recent literature review (count of referenced terms).	3
212	1-2	AI and machine learning in medical imaging brief timeline.	4
213	1-3	Example of a histopathological image segmented by multiple	
214		annotators, illustrating variations in label assignment.	7
215	1-4	Summary diagram for problem Statement	10
216	1-5	Proposed framework for the approach in [López-Pérez et al., 2024]. .	17
217	2-1	Overview of digital image concepts and their mathematical	
218		representations. The figure shows the main types of digital images	
219		(grayscale, color, multispectral, and volumetric), their mathematical	
220		representations, and the fundamental processes of sampling and	
221		quantization. Example images are included to illustrate each type. .	30
222	2-2	Histology evolution timeline. (Image from [Mazzarini et al., 2021]). .	31
223	2-3	(Above) Whole slide imaging system by Omnyx for slide	
224		digitization. (Below) Comprehensive digital pathology interface	
225		from Omnyx designed to streamline pathologists' diagnostic	
226		workflow. (From [Farahani et al., 2015]).	32
227	2-4	Common learning paradigms.	35
228	2-5	Comparative Trends of the top two most popular Deep Learning	
229		Frameworks, apparently, tendency was switched to PyTorch since	
230		2022	38

231	2-6	Annotations in the Oxford-IIIT Pet data. From left to right: pet	
232		image, head bounding box, and trimap segmentation (blue:	
233		background region; red: ambiguous region; yellow: foreground	
234		region).	39
235	2-7	Different staining techniques obtained from multi-stain breast	
236		cancer dataset [Weitz et al., 2023]. (a) shows H&E, (b) ER, (c) HER2,	
237		(d) Ki67 and (e) PGR. (f) shows an example of a Whole Slide Imaging	
238		(WSI) that was excluded since it contains multiple tissue sections. .	41
239	2-8	Screenshot of the DSA and HistomicsTK web interface while creating	
240		the crowdsourced annotations for the dataset presented by [Amgad	
241		et al., 2019].	42
242	3-1	Synthetic dataset overview for classification task	54
243	3-2	Classification results for the Correlated Chained Gaussian Processes	
244		for Multiple Annotators (CCGPMA) model using GPflux.	55
245	3-3	Ground truth and noisy labels emulation for annotators across the	
246		feature space for regression task.	55
247	3-4	Regression results for the CCGPMA model using GPflux.	56
248	4-1	Working mechanism of the proposed Truncated Generalized Cross	
249		Entropy for Semantic Segmentation ($TGCE_{SS}$) loss function. The loss	
250		combines reliability maps Λ_r with the model's predictions $f(\mathbf{X}; \theta)$	
251		to compute a weighted loss that accounts for annotator reliability	
252		across different image regions.	66
253	5-1	Original U-NET architecture.	68
254	5-2	Overview of the proposed model architecture.	69

LIST OF FIGURES

255	5-3	Training process of the proposed model overview. The model first	
256		undergoes hyperparameter optimization using Bayesian	
257		optimization, followed by end-to-end training with the optimal	
258		parameters. Estimated segmentation and reliability maps are	
259		computed for each input image, and the loss function is computed	
260		for the entire batch. Network parameters are tuned based on the	
261		loss function as objective (with minimization target) whereas the	
262		hyperparameters ϕ tuning are based on the Dice coefficient of the	
263		validation data set (with maximization target).	74
264	5-4	Example of a noisy mask generated by naively introducing random	
265		noise into a ground truth mask. Morphological consistency is lost. .	76
266	5-5	Noisy mask generated by enhancing the disturbances in the	
267		encoder layers weights for the Oxford-IIIT Pet Dataset.	
268		Morphological consistency is preserved. From left to right, SNR	
269		levels of noise in the encoder layer are 10, 5, 2, 0, -5 dB. Not adding	
270		noisy to the encoder layer would reach to an annotator behavior as	
271		good as the originally trained base model.	77
272	5-6	Example of missing labels and noisy annotations emulation for the	
273		Oxford-IIIT Pet Dataset. The same label rate ($r_l = 0.7$) is preserved	
274		for all labelers, which means in this case that every labeler is	
275		expected to label 70% of the samples, and skip the remaining 30%. .	81
276	5-7	Example evaluation of the proposed model in its last trained epoch.	
277		An accurate segmentation output is delivered as well as scalar	
278		reliability estimations from the labelers masks. Disturbance	
279		signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best	
280		found hyperparameters: learning rate = 0.0011, $q = 0.4$, noise	
281		tolerance = 0.2, $\alpha_{reg} = 0.26$, $\alpha_{entropy} = 0.03$, $\alpha_{sum} = 0.27$	82

282	5-8	Example evaluation of the proposed model in some given epoch	
283		(5/50) of training, which demonstrates a quick convergence of the	
284		model for correctly estimating the reliability of the labelers masks.	
285		An accurate segmentation output is delivered as well as pixel-wise	
286		reliability estimations from the labelers masks. Disturbance	
287		signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best	
288		found hyperparameters: learning rate = 0.0027, $q = 0.9$, noise	
289		tolerance = 0.2, $\alpha_{reg} = 0.38$, $\alpha_{entropy} = 0.02$, $\alpha_{sum} = 0.42$	82

LIST OF TABLES

291 **1-1** Summary of state-of-the-art approaches for handling multiple
292 annotators in medical image segmentation 22

293 **3-1** Comparison of Multiple Annotator Learning Methods 58

294 **5-1** Performance comparison of methods using Dice and Jaccard metrics
295 for the Oxford-IIIT Pet Dataset with no labelers aggregation. STAPLE
296 outperforms all other methods, followed by TGCE_{SS} scalar and pixel.
297 MV performs the worst. 79

LIST OF ALGORITHMS

5-1	Multi-Annotator Dataset Generation with Noisy and Missing Labels .	78
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ABBREVIATIONS

301	CAD	Computer-Aided Diagnosis 2, 5, 6
302	CCGP	Correlated Chained Gaussian Processes 18
303	CCGPMA	Correlated Chained Gaussian Processes for Multiple Annotators xviii, 17, 19,
304		53, 55, 56, 60, 82
305	CE	Cross Entropy 62, 65
306	CGP	Chained Gaussian Processes 18
307	CKA	Centered Kernel Alignment 44
308	CNN	Convolutional Neural Networks 3, 14, 24, 25, 67, 83
309	CT	Computed Tomography 1, 2, 11
310	ELBO	Evidence Lower Bound 18
311	GCE	Generalized Cross Entropy 62, 65
312	GCECDL	Generalized Cross-Entropy-based Chained Deep Learning 20
313	ISS	Image Semantic Segmentation 2, 3, 6, 11, 13, 21, 23–25, 37, 81, 82
314	KAAR	Kernel Alignment-Based Annotator Relevance Analysis 43, 44
315	LF	Latent Function 18
316	MAE	Mean Absolute Error 62, 65
317	MITs	Medical Imaging Techniques 1
318	ML	Machine Learning 11, 21
319	MRI	Magnetic Resonance Imaging 1, 2
320	MV	Majority Voting 11, 12
321	OCR	Optical Character Recognition 11
322	PET	Positron Emission Tomography 14
323	ROI	Region of Interest 2, 6, 15, 33
324	SLFM	Semi-Parametric Latent Factor Model 18, 51, 53
325	SS	Semantic segmentation 3
326	STAPLE	Simultaneous Truth and Performance Level Estimation 12–14, 38
327	WSI	Whole Slide Imaging xviii, 1, 5, 6, 8, 15, 31, 41

328

329

CHAPTER

330

ONE

331

332

INTRODUCTION

333

1.1 Motivation

334 Since Roentgen's discovery of X-rays in 1895, medical imaging has advanced
335 significantly, with modalities like radionuclide imaging, ultrasound, **Computed**
336 **Tomography (CT)**, **Magnetic Resonance Imaging (MRI)**, and digital radiography
337 emerging over the past 50 years. Modern imaging extends beyond image
338 production to include processing, display, storage, transmission and analysis.
339 [Zhou et al., 2021]. Other **Medical Imaging Techniques (MITs)** have arose during
340 the last decades, some of them implying only the examination of certain pieces or
341 tissues instead of complete patients, like histopathological images, which are
342 images of tissue samples obtained from biopsies or surgical resections and are
343 widely used for the diagnosis of diseases like cancer through **Whole Slide Imaging**
344 (**WSI**) scanners [Rashmi et al., 2021].

345 Along with the advances in technologies for medical images acquisition,
346 computational technologies on pattern recognition and artificial intelligence have

also emerged, allowing the development of **Computer-Aided Diagnosis (CAD)** systems based on machine learning algorithms. These systems aim to assist physicians in the diagnosis and treatment of diseases, by providing a second opinion or by automating the analysis of medical images. [Panayides et al., 2020]. One of the most used tasks in which machine learning technologies is being used in the universe of medical images is **Image Semantic Segmentation (ISS)**, which consists of assigning a label to each pixel in an image according to the object it belongs to. This task is crucial for the development of **CAD** systems, as it allows the identification of **Region of Interest (ROI)** in the images, which can be used to detect and classify diseases [Azad et al., 2024].

The application of Machine Learning in medical imaging has grown significantly, with key tasks including classification, segmentation, anomaly detection, super-resolution, image registration, and synthetic image generation [Brito-Pacheco et al., 2025]. Among imaging modalities, X-rays and **CT** scans are widely used for classification and anomaly detection, especially in pulmonary and oncological applications. **MRI** and ultrasound play a crucial role in segmentation and resolution enhancement, while PET/SPECT imaging is essential for anomaly detection in oncology and neurodegenerative diseases [Brito-Pacheco et al., 2025]. Histopathology is rapidly gaining prominence, particularly in segmentation and feature extraction, where AI-driven techniques aid in automated cancer diagnosis and tissue structure analysis. The integration of Deep Learning in histological image processing is revolutionizing pathology, enabling more precise and efficient diagnostics. A brief comparison of the tasks and medical image types based on recent literature review, can be seen in Figure 1-1. [Yu et al., 2025], [Brito-Pacheco et al., 2025], [Ryou et al., 2025], [Hu et al., 2025], [Elhaminia et al., 2025]

For solving the different requirements of tasks in medical images, a variety of computational techniques have been developed [Zhou et al., 2021]. Initially, these needs were covered with simple morphological filters, which implied no training process or elaborated optimization. However, as the complexity of the tasks increased, the need for more sophisticated techniques arose, leading to the

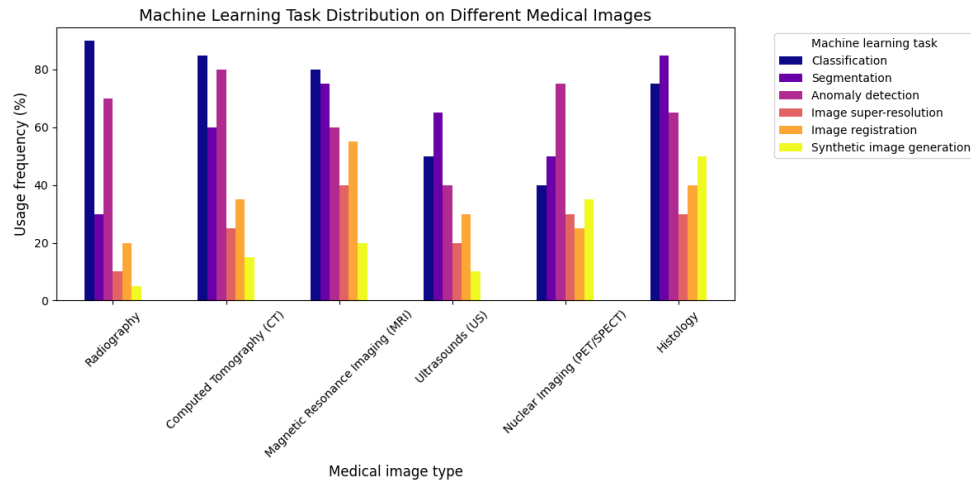


Figure 1-1 Estimation of the popularity of tasks and medical image types based on recent literature review (count of referenced terms).

377 application of advanced statistical tools and machine learning algorithms like
 378 Support Vector Machines, Decision Trees, and SGD Neural Networks [Avanzo
 379 et al., 2024]. The coevolution of advances in medical image acquisition,
 380 computational power (i.e. Moore's law) and statistical/mathematical techniques
 381 have led to a convergence for merging state of the art algorithms with medical
 382 imaging [Shalf, 2020]. Figure 1-2 shows a brief timeline of coevolution between
 383 some conspicuous advances in computational pattern recognition and its medical
 384 applications in different scopes (besides medical imaging) [Avanzo et al., 2024].

385 Convolutional Neural Networks (CNN) have been widely used in Semantic
 386 segmentation (SS) tasks, as they have outperformed traditional machine learning
 387 algorithms in this task for both medical and non medical images [Xu et al., 2024]
 388 [Sarvamangala and Kulkarni, 2022]. However, most CNN architectures are deep,
 389 which imply a necessity of a large amount of data to train them. This introduces a
 390 problem since both the acquisition and annotation of medical images are
 391 expensive and time-consuming processes. This is especially true for ISS tasks, as
 392 they require pixel-level annotations, which is taxing in terms of cost, time and
 393 logistics involved [Bhalgat et al., 2018]. Other fashions face this problem through
 394 less expensive annotation strategies like bounding boxes or anatomical landmarks

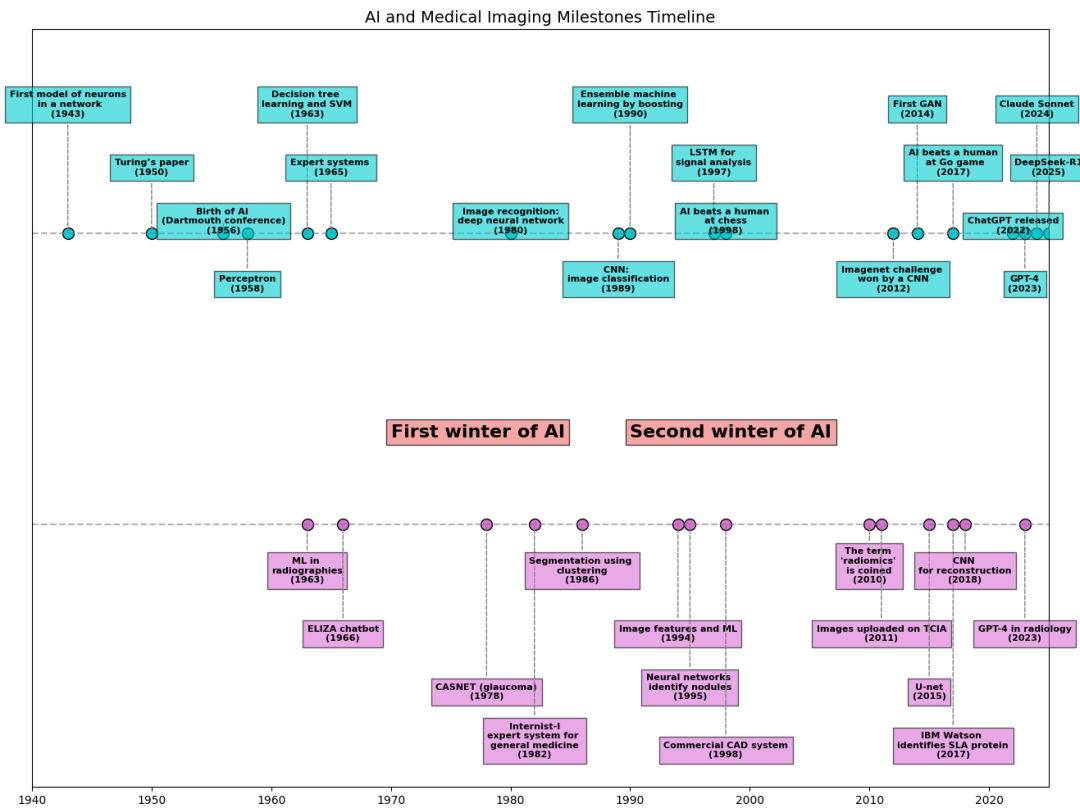


Figure 1-2 AI and machine learning in medical imaging brief timeline.

395 for being used in a semi-supervised strategy [Shah et al., 2018].

396 Many medical images datasets however, contain a high variability in class sizes
397 and variations in colors, which is specially noticeable in histopathological images
398 because of the usage of different staining and other factors which can affect the
399 color of the images (see section 2.2.3). This variability can lead to a significant loss
400 of efficiency of machine learning models when using a mixed supervision strategy,
401 as the model can be biased towards the most common classes or colors in the
402 dataset [Shah et al., 2018].

403 This is where other solutions arise to tackle the problem of the weak image
404 annotation while maintaining low costs. One of these solutions is crowdsourcing
405 strategy, which consists of having multiple annotators labeling the same image,
406 and then combining the labels to obtain a consensus label [Lu et al., 2023]. This
407 strategy can lead to a labeling cost reduction when different levels of expertise are
408 combined, since the crowd may be composed of both experts and laymen, being
409 the latter less expensive to hire [López-Pérez et al., 2023].

410 Recently, diagnosis, prognosis and treatment of cancer have heavily relied on
411 histopathology, where tissue samples are obtained through biopsies or surgical
412 resections and critical information that helps pathologists determine the presence
413 and severity of malignancies [López-Pérez et al., 2024]. The segmentation of
414 histopathological images enables precise identification of structures such as
415 nuclei, glands, and tumors, which are essential for assessing disease progression
416 and treatment response [Rashmi et al., 2021]. Accurate segmentation is
417 particularly crucial in digital pathology, where whole-slide images (WSI) are
418 analyzed using AI-powered CAD systems to support clinical decision-making
419 [López-Pérez et al., 2024].

420 A major challenge in histopathological image segmentation arises from the
421 variability in annotations provided by different pathologists. Unlike natural
422 images, where object boundaries are often well-defined, histological structures
423 may have ambiguous borders, leading to inconsistencies among annotators

[López-Pérez et al., 2023]. Because of this, crowdsourcing labeling is one of the most popular approaches, as illustrated in Figure 1-3, an example of how histopathological images are segmented by multiple experts, showing some variations in label assignment ¹. These discrepancies highlight the need for models that can handle annotation uncertainty effectively. Leveraging crowdsourcing strategies and machine learning techniques that infer annotator reliability can enhance segmentation performance while reducing costs.

1.2 Problem Statement

Throughout the development of medical technology and CAD, the task of ISS has become a crucial step in delivering precise diagnosis and treatment planning [Giri and Bhatia, 2024]. Particularly, in the area of histopathological studies, the usage of Whole Slide Images (WSI) is rather common since this method delivers high quality imaging and allows for the diagnosis of diseases like cancer [Lin et al., 2024].

ISS task consists of assigning a label to each pixel in an image according to the object it belongs to. Accurate segmentation is essential for the development of CAD systems, as it allows the identification of regions of interest (ROI) in the images, which can be used to detect and classify diseases and hence, treatment planning [Sarvamangala and Kulkarni, 2022]. However, modern computational solutions for ISS tasks involve the use of deep learning, which mostly rely large amounts of labeled data to train the models on supervised learning techniques. This means that the model is trained on a dataset with ground-truth labels, which are assumed to be correct and consistent across all samples. In practice, this assumption is often violated due to the high technical complexity of labeling these segments ².

¹obtained from a real world Triple Negative Breast Cancer (TNBC) dataset published in [López-Pérez et al., 2023]

²compared to a more trivial task like image classification on ordinary an well known classes like MNIST

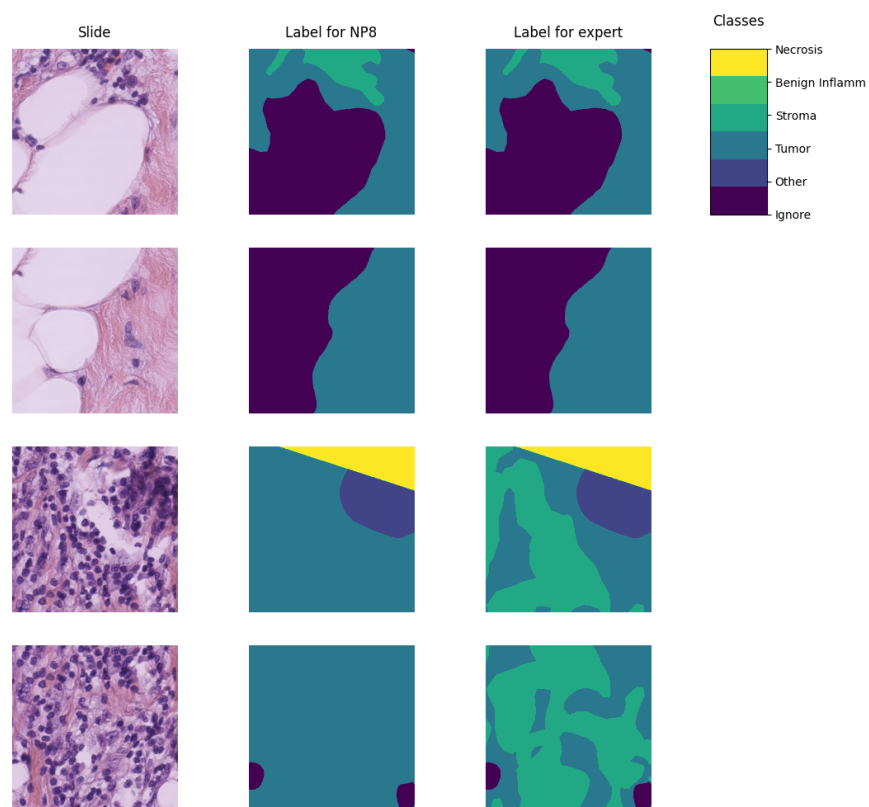


Figure 1-3 Example of a histopathological image segmented by multiple annotators, illustrating variations in label assignment.

The process of labeling medical images is often managed with the help of specialized software tools that allow the annotators to draw the regions, delivering an standard format for the labeled masks [Habis, 2024]. Despite the help of these tools, the labeling process in WSI can have high costs, as it requires long hours of work from specialized personnel. Because of cost constraints in many medical institutions, the labeling processes is often done by multiple labelers with varying levels of expertise, equalizing the cost of the labeling process. However, this strategy can lead to inconsistent labels, as the consensus between the labelers may not be exact due to the diversity in depth of knowledge and experience of the labelers [Xu et al., 2024]. These inconsistencies are mostly represented in the subsections 1.2.1 and 1.2.2.

1.2.1 Variability in Expertise Levels

One of the primary sources of inter-observer variability in medical image segmentation is the difference in expertise levels among annotators [López-Pérez et al., 2023]. Experienced radiologists and pathologists tend to produce highly precise annotations, whereas novice labelers may introduce systematic biases due to their limited familiarity with subtle image features. Studies have demonstrated that annotation accuracy tends to improve with experience, yet medical institutions often rely on a mix of annotators to manage costs and workload distribution [Lu et al., 2023].

The training background of annotators and institutional guidelines play a crucial role in shaping labeling practices. Different medical schools and hospitals may adopt distinct segmentation protocols, leading to inconsistencies when datasets are combined from multiple sources [López-Pérez et al., 2023]. For example, some institutions may emphasize conservative delineation of tumor boundaries, while others adopt a more inclusive approach. Such variations contribute to systematic biases in medical image datasets [Banerjee et al., 2025].

Medical images frequently contain structures with ambiguous boundaries, making segmentation inherently subjective. For instance, tumor margins in histopathological slides may not have well-defined edges, leading to variations in how different annotators delineate the regions of interest [Carmo et al., 2025]. These discrepancies arise not only from technical expertise but also from differences in perception and interpretation.

1.2.2 Technical Constraints and Image Quality

Technical constraints in medical imaging, such as resolution differences, noise levels, and contrast variations, can significantly impact segmentation accuracy. Lower-resolution images may obscure fine structures, leading to inconsistencies in boundary delineation [Zhou et al., 2024].

When combined with long sessions, bad images might also increase the cognitive load of the annotators, leading to fatigue and reduced precision in labeling [Kim et al., 2024]. This is particularly relevant in histopathological studies, where the staining process and tissue preparation can introduce color variations and artifacts that affect image quality, even if the same scanning equipment is used [Karthikeyan et al., 2023].

1.2.3 Research Question

Given the challenges posed by inconsistent labels in medical image segmentation, this work aims to address the following research question:

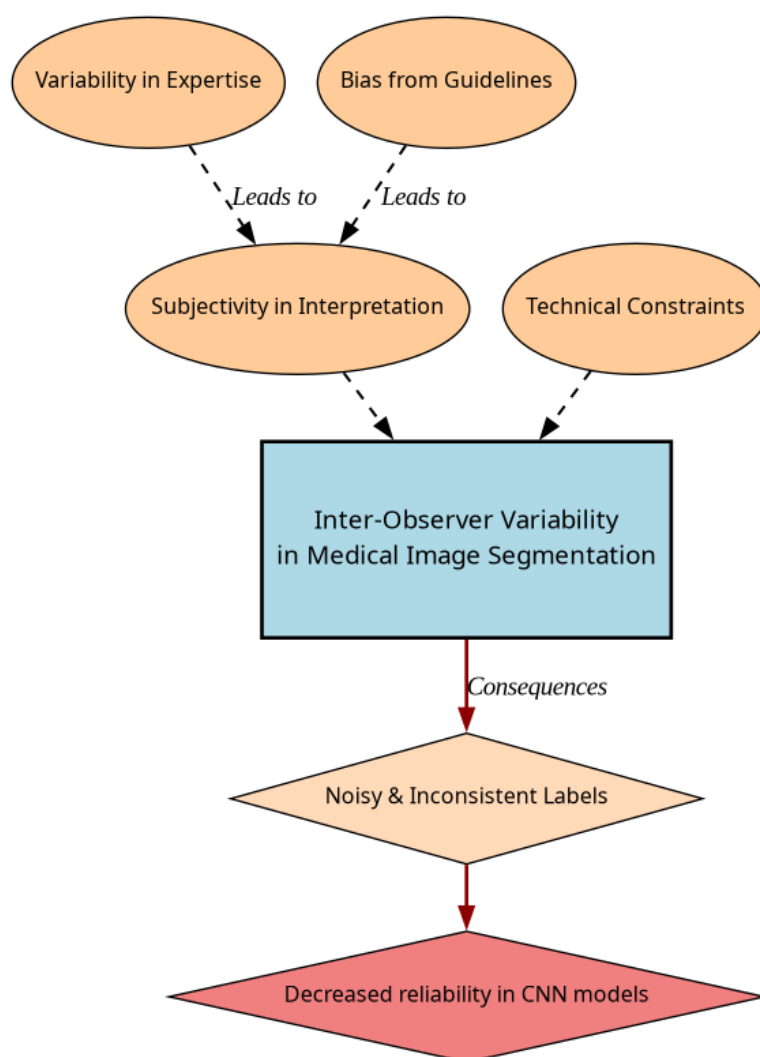


Figure 1-4 Summary diagram for problem Statement

Research Question

How can we develop a learning approach for ISS tasks in medical images that can adapt to inconsistent labels without requiring explicit supervision of labeler performance, while addressing challenges related to variability in expertise levels and technical constraints, and maintaining interpretability, generalization, and computational efficiency?

495

1.3 Literature Review

496

Certainly, in general Machine Learning (ML) classification tasks³ where multiple annotators are involved, Majority Voting (MV) is by far the simplest possible approach to implement. This concept was born multiple times and divergently in multiple fields, but it was described as relevant for ML and pattern recognition labeling for classification in [Lam and Suen, 1997], in which the approach is exposed as simple, yet powerful. The authors describe the MV as a method that can be used to improve the accuracy of classification tasks by combining the labels of multiple annotators. The method is based on the assumption that the majority vote of the annotators is more likely to be correct than the vote of a single annotator. The authors also describe the method as a straightforward way to improve the accuracy of classification tasks without the need for complex algorithms or additional data. The authors also prove this method to deliver very similar results to more complicated approaches (Bayesian, logistic regression, fuzzy integral, and neural network) in the particular task of Optical Character Recognition (OCR). Despite its simplicity, modern solutions for delivering accurate medical image segmentation models still rely on Majority Voting at some stage, like [Elnakib et al., 2020], which uses a majority voting strategy for delivering a final output based on the labels of multiple models (VGG16-Segnet, Resnet-18 and Alexnet) in CT images for Liver Tumor Segmentation, or [López-Pérez et al., 2023],

³In this work, image segmentation is considered as a particular case of classification in which target classes are assigned pixel-wise.

which uses MV for combining noisy annotations as an additional annotator to be included in the deep learning solution. Majority voting as a technique for setting a pseudo ground truth label is a powerful approach for its simplicity in many use cases in which the target to be labeled is not tied to an expertise related task, otherwise, the assumption of equal expertise among the labelers can be a source of bias in the final label, which is not desirable in the case of highly technical annotations like medical images. In subsection 1.3.1, we will be reviewing literature which no longer assumes the naive approach of equal expertise among labelers and face the challenge of learning from inconsistent labels.

1.3.1 Facing Annotation Variability in Medical images

Learning from crowds approaches in general face the challenge of not having a ground truth label and hence, an intrinsic difficulty in measuring the real reliability of the labelers annotations. Some approaches assume beforehand a certain level of expertise for each labeler based on experience as an input, like in [TIAN and Zhu, 2015], which introduce the concept of max margin majority voting, using the reliability vector as weights for the weights for the binary and multiclass classifier. The crowdsourcing margin is the minimal difference between the aggregated score of the potential true label and the scores for other alternative labels. Accordingly, the annotators' reliability is estimated as generating the largest margin between the potential true labels and other alternatives. The problem introduced in this approach is assuming an stationary reliability per expert across the whole input space, which is imprecise since annotators performance may change between different tasks or even between different regions of the same image.

STAPLE Mechanism

The Simultaneous Truth and Performance Level Estimation (STAPLE) algorithm, introduced in [Warfield et al., 2004] is a probabilistic framework that estimates a

hidden true segmentation from multiple segmentations provided by different raters. It also estimates the reliability of each rater by computing their sensitivity and specificity.

The **STAPLE** algorithm's goal is to maximize the log likelihood function:

$$(\mathbf{p}, \mathbf{q}) = \arg \max_{\mathbf{p}, \mathbf{q}} \ln f(\mathbf{D}, \mathbf{T} \mid \mathbf{p}, \mathbf{q}). \quad (1-1)$$

Where $\mathbf{D}$ is the set of segmentations provided by the raters, $\mathbf{T}$ is the hidden true segmentation, p is the sensitivity and q is the specificity of the raters.

This is achieved by using the Expectation-Maximization algorithm to maximize the log likelihood function in equation, which is done iteratively with step computations:

$$\begin{aligned} (p_j^{(k)}, q_j^{(k)}) = \arg \max_{p_j, q_j} \sum_{i: D_{ij}=1} W_i^{(k-1)} \ln p_j \\ + \sum_{i: D_{ij}=1} \left(1 - W_i^{(k-1)}\right) \ln(1 - q_j) \\ + \sum_{i: D_{ij}=0} W_i^{(k-1)} \ln(1 - p_j) \\ + \sum_{i: D_{ij}=0} \left(1 - W_i^{(k-1)}\right) \ln q_j. \end{aligned} \quad (1-2)$$

The capacity of STAPLE to accurately estimate the true segmentation, even in the presence of a majority of raters generating correlated errors, was demonstrated, which makes it theoretically a strong choice for setting a ground-truth in binary or multiclass medical **ISS** tasks.

The popularity and performance of **STAPLE** has led to its usage in modern applications medical image, 3d spatial images due to its assumption of decision

space being based on voxel-wise decisions, like the authors in [Grefve et al., 2024] which applied the algorithm on Positron Emission Tomography (PET) images. Other authors still rely heavily on STAPLE for setting a ground truth consensus for histopathological images, like [Qiu et al., 2022].

However, the STAPLE algorithm has some limitations. It assumes independent rater errors, which may not hold in practice, leading to biased estimates. STAPLE is also sensitive to low-quality annotations, potentially degrading final segmentations if the weights are not initialized correctly. The algorithm tends to over-smooth results, blurring fine details, and struggles with multi-class segmentation. Computationally, it is expensive due to its iterative EM approach. Additionally, STAPLE cannot correct systematic biases in annotations and depends on initial estimates, impacting accuracy. Lastly, the estimated performance levels lack interpretability, making it difficult to assess annotator reliability effectively.

Finally, this work contemplates STAPLE as useful for label aggregation, hence being a good support for other methods, but not that useful for providing annotations of structures on new and unlabeled images.

U-shaped CNNs

Since the introduction of U-Net [Ronneberger et al., 2015] in 2015 for biomedical image segmentation, U-shaped CNNs have become a prevalent architecture in medical image segmentation tasks. The U-Net’s success stems from its ability to capture both global and local information through its contracting and expanding paths, making it particularly effective for complex and heterogeneous structures, even with limited annotated data. This architecture has been successfully applied to various medical image segmentation tasks, including organ segmentation, tumor segmentation, and brain structure segmentation.

The U-Net architecture consists of a symmetric encoder-decoder structure with skip connections. The encoder path progressively reduces spatial dimensions

while increasing feature channels through a series of convolutional and max-pooling layers, capturing high-level semantic information. The decoder path uses transposed convolutions to gradually recover spatial resolution while reducing feature channels. Skip connections between corresponding encoder and decoder layers preserve fine-grained details by concatenating high-resolution features from the encoder with upsampled features in the decoder, enabling precise localization of structures.

U-Net based approaches

In [López-Pérez et al., 2024] two networks are trained for delivering a final segmentation. One network is trained to estimate the annotators reliability and another one is trained to segment the image. The first network is a deep neural network that takes as input features of image and the labelers id encoded as one-hot and outputs a reliability map across the image feature space. This map is then used to weight the contribution of each annotator to the final segmentation. The second network is the U-Net used for segmentation.

In this approach, it is assumed that the images are labeled for at least one labeler and not all of them, which is closer to a real world scenario, in which it is common to have images with variability in the amount of annotations, per patch. Hence, the input data can be modeled as:

$$\mathcal{D} = (\mathbf{X}, \tilde{\mathbf{Y}}) = \{(\mathbf{x}_n, \tilde{\mathbf{y}}_n^r) : n = 1, \dots, N; r \in R_n\}, \quad (1-3)$$

Where every $\mathbf{x}_n$ is an input patch from a ROI in one WSI, $\tilde{\mathbf{y}}_n$ is the noisy annotation from the r labeler, N is the number of patches in the dataset and $R_n \subset \{1, \dots, R\}$ is the set of labelers that annotated the image $\mathbf{x}_n$.

The authors then assume the annotator network to deliver a reliability map $\{\hat{\mathbf{A}}_\phi^{(r)}(\mathbf{x})\}_{r \in R_n}$ with different dimensions:

- 608 • CR global: a single reliability vector per labeler with dimensions C which
609 represent global reliability of the labeler across all input space.
- 610 • CR image: a single reliability vector per image per labeler with dimensions C
611 which represent local reliability of the labeler across the image.
- 612 • CR pixel: a reliability matrix per image per labeler, with dimensions C which
613 represent local reliability of the labeler across all the pixels in the image.

614 These differences in dimensions are determined by the feature extraction space
615 from segmentation network which feed the input of the annotator network, which
616 the authors vary for experimentation purposes.

617 Being $\mathbf{p}_\theta(\mathbf{x}_n)$ the estimation of the latent (ground truth) segmentation delivered by
618 the segmentation UNet network, thus, the estimated segmentation probability
619 mask for each annotator is given by the product:

$$\mathbf{p}_{\theta,\phi}^{(r)}(\mathbf{x}_n) := \mathbf{A}_\phi^{(r)}(\mathbf{x}) \odot \mathbf{p}_\theta(\mathbf{x}_n), \quad (1-4)$$

620 where $\odot$ is the element-wise product and ϕ and θ are the parameters of the
621 annotator network and the segmentation UNet network, respectively, being the
622 latter initialized with a ResNet34 backbone pre-trained on ImageNet.

623 The authors propose a loss function involving cross-entropy and a trace based
624 regularization on the reliability map, originally proposed in [Zhang et al., 2020]
625 which combined, looks like:

$$\mathcal{L}(\theta, \phi) := \sum_{n=1}^N \sum_{r=1}^R \mathbb{I}(\tilde{\mathbf{y}}_n^{(r)} \in R_n) \cdot \left[\text{CE} \left(\mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \cdot \mathbf{p}_\theta(\mathbf{x}_n), \tilde{\mathbf{y}}_n^{(r)} \right) + \lambda \cdot \text{tr} \left(\mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \right) \right] \quad (1-5)$$

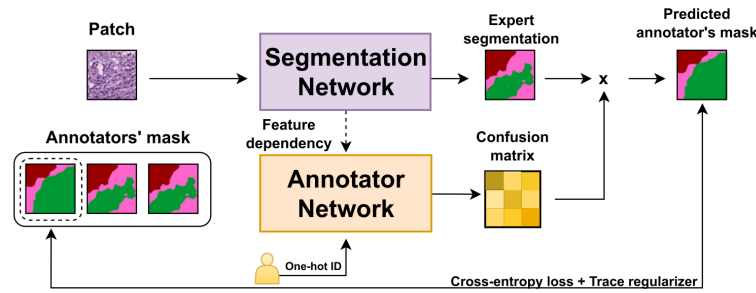


Figure 1-5 Proposed framework for the approach in [López-Pérez et al., 2024].

Being $\mathbb{I}$ the indicator function, CE the cross-entropy loss, and λ the regularization parameter.

When evaluated on a Triple Negative Breast Cancer dataset, this approach achieves a Dice coefficient of 0.7827, outperforming STAPLE (0.7039) and matching expert-supervised performance (0.7723). The CR image reliability modeling proved most effective, as CR pixel, while potentially offering finer-grained reliability estimation, requires significantly more training data.

Despite the decent performance of the approach, solving the problem of multiple labelers with two networks can be overwhelming for the optimization process, requiring large amounts of annotated data to properly codify the annotators spatial reliabilities, which could be managed by a single model with an appropriate loss function.

Bayesian models

Bayesian approaches are a good choice for handling label noise and uncertainty in the labelers. In [Julián and Álvarez Meza Andrés Marino, 2023] the authors propose a novel approach from Gaussian Processes to model the relationship between the annotators' reliability and the input data, while also preserving the interdependencies among the annotators. This is achieved by introducing Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA), a

framework based on the well known **Chained Gaussian Processes (CGP)**. CGP on itself cannot consider inter-annotator dependencies, thus, the authors introduce the **Correlated Chained Gaussian Processes (CCGP)** to model correlations between the GP latent functions, which are supposed to be generated from a **Semi-Parametric Latent Factor Model (SLFM)**:

$$f_j(\mathbf{x}_n) = \sum_{q=1}^Q w_{j,q} \mu_q(\mathbf{x}_n), \quad (1-6)$$

where $f_j : \mathcal{X} \rightarrow \mathbb{R}$ is a **Latent Function (LF)**, $\mu_q(\cdot) \sim \mathcal{GP}(0, k_q(\cdot, \cdot))$ with $k_q : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ being a kernel function, and $w_{j,q} \in \mathbb{R}$ is a combination coefficient ($Q \in \mathbb{N}$). This leads to a joint distribution of the form:

$$p(\mathbf{y}, \hat{\mathbf{f}}, u | \mathbf{X}) = p(\mathbf{y} | \boldsymbol{\theta}) \prod_{j=1}^J p(\mathbf{f}_j | \mathbf{u}) p(\mathbf{u}), \quad (1-7)$$

where $\mathbf{y}$ is the vector of noisy labels, $\hat{\mathbf{f}}$ is the vector of latent functions, u represents the inducing points, and $\mathbf{X}$ is the input data.

Combined with inducing-variables based methods for sparse GP approximations, and maximizing an **Evidence Lower Bound (ELBO)** for the estimation of the variational parameters, the authors reach a model whose variational expectations are not analytically tractable, and hence, the authors derive a Gaussian-Hermite quadrature approach.

Finally, the authors extend this approach for being applied to classification and regression, reaching the only known approach to involve chained gaussian processes in multiple annotators classification and regression tasks while

663 preserving the interdependencies among the annotators, and also outperforming
 664 GPC-MV⁴, MA-LFC-C⁵, MA-DGRL⁶, MA-GPC⁷, MA-GPCV⁸, MA-DL⁹, KAAR¹⁰.

665 CCGPMA on itself proposes a good approach for handling label noise and
 666 uncertainty in the labelers for regression and classification tasks, while also
 667 preserving the interdependencies among the annotators, however, it does not face
 668 the image segmentation problem, which is the main focus of this works, however,
 669 it does not face the image segmentation problem, which is the main focus of this
 670 work. Besides, handling so many latent functions during the optimization process
 671 is computationally expensive, making it on itself infeasible for large and high
 672 resolution datasets.

673 1.3.2 Facing Noisy Annotations and Low-Quality Data

674 The problem of low-quality data and noisy annotations has been tackled with
 675 various strategies. One such approach is the use of deep learning models that
 676 incorporate loss functions designed to mitigate the effects of unreliable labels.
 677 Traditional methods such as Majority Voting (MV) or Expectation-Maximization
 678 (EM) have been widely used for aggregating multiple annotators' inputs. However,
 679 they assume a homogeneous reliability of annotators, which may not hold in
 680 real-world scenarios.

⁴A GPC using the MV of the labels as the ground truth.

⁵A LRC with constant parameters across the input space.

⁶A multi-labeler approach that considers as latent variables the annotator performance.

⁷A multi-labeler GPC, which is an extension of MA-LFC.

⁸An extension of MA-GPC that includes variational inference and priors over the labelers' parameters.

⁹A Crowd Layer for DL, where the annotators' parameters are constant across the input space.

¹⁰A kernel-based approach that employs a convex combination of classifiers and codes labelers dependencies.

Loss functions in deep learning models

Loss functions are fundamental components in deep learning models that quantify how well a model's predictions match the ground truth. They serve as the objective function that guides the learning process by measuring the discrepancy between predicted and actual values. In classification tasks, the most common loss functions are Cross-Entropy (CE) and Mean Absolute Error (MAE) [Zhang and Sabuncu, 2018]. CE is particularly effective for classification as it heavily penalizes confident but wrong predictions, though it can be sensitive to noisy labels. MAE, on the other hand, is more robust to outliers and assigns equal weights to all mistakes, but typically requires more training iterations. For image segmentation tasks, specialized loss functions have been developed to handle the unique challenges of pixel-wise classification. The Dice loss, which measures the overlap between predicted and ground truth regions, is widely used in medical image segmentation [Zhao et al., 2020]. More recently, the Generalized Cross Entropy (GCE) loss has emerged as a robust alternative that combines the benefits of both CE and MAE, allowing for better handling of noisy labels through a tunable parameter that controls sensitivity to outliers. However, to the best of our knowledge, none of the current loss functions approaches have focused on the multiple labelers scenario, neither on the per labeler spatial reliability estimation.

Generalized Cross-Entropy for multiple annotators classification

A more recent approach was proposed by [Triana-Martinez et al., 2023], introducing a Generalized Cross-Entropy-based Chained Deep Learning (GCECDL) framework. This method addresses the limitations of traditional label aggregation techniques by modeling each annotator's reliability as a function of the input data. The approach effectively mitigates the impact of noisy labels by using a noise-robust loss function, balancing Mean Absolute Error (MAE) and Categorical Cross-Entropy (CE). Unlike prior approaches, GCECDL accounts for the dependencies among annotators while encoding their non-stationary behavior

709 across different data samples. Their experiments on multiple datasets
 710 demonstrated superior predictive performance compared to state-of-the-art
 711 methods, particularly in cases where annotations were highly inconsistent.

712 The strategy of the authors effectively unlocks the potential of **ML** models to handle
 713 low-quality data and noisy annotations, but it is bounded to classifications tasks
 714 only, not being by itself applicable to segmentation tasks. The TGCE equation for
 715 handling multiple annotators is defined as:

$$\text{TGCE}(\mathbf{y}, f(\mathbf{x}); \tilde{\lambda}_x, \tilde{C}) = \tilde{\lambda}_x \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q} + (1 - \tilde{\lambda}_x) \frac{1 - (\tilde{C})^q}{q}, \quad (1-8)$$

716 where $\tilde{\lambda}_x$ represents the annotator reliability, $\tilde{C}$ is a constant, q is a parameter that
 717 controls the balance between MAE and CE behavior, $\mathbf{y}$ is the annotation vector, and
 718 $f(\mathbf{x})$ is the model prediction. This approach is more deeply discussed in chapter 4.

719 1.4 Aims

720 1.4.1 General Aim

721 The main purpose of this work is to develop a novel approach for **ISS** tasks in
 722 medical images, which can adaptively infer the best possible segmentation without
 723 needing separate inputs about the labeler performance. This approach is expected
 724 to properly offer a much more interpretable reliability space per labeler than other
 725 solutions in the literature, correctly facing the labeler performance inconsistency
 726 across the annotators space and the variability of images quality.

Table 1-1 Summary of state-of-the-art approaches for handling multiple annotators in medical image segmentation

Approach	Key Characteristics	Advantages	Limitations
Majority Voting (MV)	Simple aggregation of multiple annotators' labels	- Simple to implement - No complex algorithms required - Proven effective in many cases	- Assumes equal expertise - Sensitive to correlated errors - May introduce bias in expert tasks
STAPLE	Probabilistic framework estimating true segmentation and rater reliability	- Handles binary and multiclass segmentation - Estimates rater sensitivity/specificity - Robust to correlated errors	- Computationally expensive - Sensitive to low-quality annotations - Tends to over-smooth results - Limited interpretability
U-Net based approaches	Deep Learning architecture with encoder-decoder structure	- Captures both global and local information - Effective for complex structures - Works well with limited data	- Requires large amounts of training data - Complex optimization process - May need specialized loss functions
Bayesian (CCGPMA)	Models Gaussian Processes modeling annotator reliability	- Preserves annotator interdependencies - Handles label noise effectively - Works for classification and regression	- Computationally expensive - Not directly applicable to segmentation - Complex optimization process
Generalized Entropy (GCE)	Cross-Loss function balancing MAE and CE	- Robust to noisy labels - Accounts for non-stationary behavior - Handles annotator dependencies	- Limited to classification tasks - Requires parameter tuning - May need adaptation for segmentation

1.4.2 Specific Aims

- To develop a novel loss function for ISS tasks in medical images, capable of inferring the best possible segmentation without needing separate inputs about the labeler performance.
- Introducing a tensor map which codifies the reliability of each labeler, allowing the model to implicitly weigh the labelers based on their performance across the mask and classes space.
- To develop and test a deep learning model for ISS tasks in medical images, which can learn from inconsistent labels and improve the segmentation performance compared to other solutions in state of the art.

1.5 Outline and Contributions

Building upon the considerations discussed in Section 1.3, this thesis introduces a novel approach to ISS in medical imaging. The core of this work is a model designed to adapt its learning process to varying labeler performance. This is achieved through a unique loss function that infers optimal segmentation and assesses labeler reliability without requiring explicit performance inputs. This loss function implicitly weighs labelers based on their performance, utilizing an intermediate reliability map. This map enables the model to prioritize information from reliable labelers while disregarding noisy labels. Consequently, this methodology diverges from conventional CNN-based segmentation models by eliminating the need for direct supervision of labeler performance, thereby enhancing its generalizability and adaptability across diverse datasets and labeler cohorts.

The primary contributions of this research to the field of ISS in medical images are as follows:

- The development of a novel loss function for ISS tasks. This function can infer the best possible segmentation and codify labeler reliability without needing separate inputs regarding labeler performance or the training of multiple networks.
- The creation of a Chained Deep Learning model for ISS in medical images. This model effectively learns from inconsistent labels and demonstrates improved segmentation performance, outperforming state-of-the-art approaches on a Triple Negative Breast Cancer histopathology dataset.
- A mechanism for emulating noisy annotations. This involves using network encoding layer weight disturbances to construct a crowdsourced version of the well-known Oxford-IIIT Pet Dataset.

- 763 • A Python package facilitating the use of the proposed loss function and
764 architecture within CNN models for ISS tasks in medical images.¹¹
- 765 • Mappings of the Triple Negative Breast Cancer and Oxford-IIIT Pet Datasets as
766 lazy loaders, integrated into the package for immediate use with the proposed
767 loss function.¹²
- 768 • A publicly accessible Github repository containing the source code developed
769 for this work.¹³

¹¹https://pypi.org/project/seg_tgce/

¹²<https://seg-tgce.readthedocs.io/en/latest/experiments.html>

¹³https://github.com/blotero/seg_tgce

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CHAPTER

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TWO

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CONCEPTUAL PRELIMINARIES

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2.1 Modern Concept of Digital Image

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A digital image is a numerical representation of a visual scene, captured through various imaging devices and stored in a computer. From a mathematical perspective, a digital image can be represented as a function $f(x, y)$ that maps spatial coordinates (x, y) to intensity values. In the discrete domain, this function is sampled at regular intervals, creating a matrix of values known as pixels (picture elements).

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2.1.1 Types of Digital Images

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Grayscale Images

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Grayscale images are the simplest form of digital images, where each pixel represents a single intensity value. Mathematically, a grayscale image can be represented as a 2D matrix I of size $M \times N$, where each element $I(i, j)$ represents the intensity at position (i, j) . The intensity values typically range from 0 (black) to 255 (white) in 8-bit images, or from 0 to 65535 in 16-bit images.

788 Color images

789 Color images extend the grayscale concept by representing each pixel with multiple
790 channels, typically Red, Green, and Blue (RGB). A color image can be represented
791 as a 3D matrix I of size $M \times N \times 3$, where $I(i, j, k)$ represents the intensity of the
792 k -th color channel at position (i, j) . Other color spaces like HSV (Hue, Saturation,
793 Value) or CMYK (Cyan, Magenta, Yellow, Key) are also commonly used in different
794 applications.

795 Multispectral images

796 Multispectral images capture information across multiple wavelength bands
797 beyond the visible spectrum. These images can be represented as a 3D matrix I of
798 size $M \times N \times B$, where B is the number of spectral bands. Each band $I(i, j, b)$
799 represents the intensity at position (i, j) for the b -th spectral band. This
800 representation is particularly useful in medical imaging, remote sensing, and
801 scientific applications.

802 3D images and volumetric data

803 Three-dimensional images extend the concept of pixels to voxels (volume elements).
804 A 3D image can be represented as a 3D matrix V of size $M \times N \times D$, where D
805 represents the depth dimension. Each voxel $V(i, j, k)$ represents the intensity at
806 position (i, j, k) in the 3D space. This representation is fundamental in medical
807 imaging (CT, MRI), scientific visualization, and computer graphics.

808 2.1.2 Mathematical Representations

809 The mathematical foundation of digital images relies on several key concepts:

- 810 • **Sampling:** The process of converting a continuous image into a discrete
811 representation. According to the Nyquist-Shannon sampling theorem, the
812 sampling frequency must be at least twice the highest frequency present in
813 the image to avoid aliasing.
- 814 • **Quantization:** The process of converting continuous intensity values into
815 discrete levels. The number of quantization levels determines the image's
816 bit depth and affects its quality and storage requirements.
- 817 • **Resolution:** The number of pixels per unit length in an image, typically
818 measured in pixels per inch (PPI) or dots per inch (DPI).
- 819 • **Dynamic range:** The ratio between the maximum and minimum measurable
820 light intensities in an image, often expressed in decibels (dB).

821 The mathematical representation of a digital image can be expressed as:

$$I(x, y) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} f(i, j) \cdot \delta(x - i, y - j) \quad (2-1)$$

822 where $I(x, y)$ is the digital image, $f(i, j)$ represents the intensity values, and $\delta(x -$
823 $i, y - j)$ is the Kronecker delta function.

824 For color images, the representation extends to:

$$I(x, y) = \begin{bmatrix} I_R(x, y) \\ I_G(x, y) \\ I_B(x, y) \end{bmatrix} \quad (2-2)$$

825 where I_R , I_G , and I_B represent the red, green, and blue channels respectively.

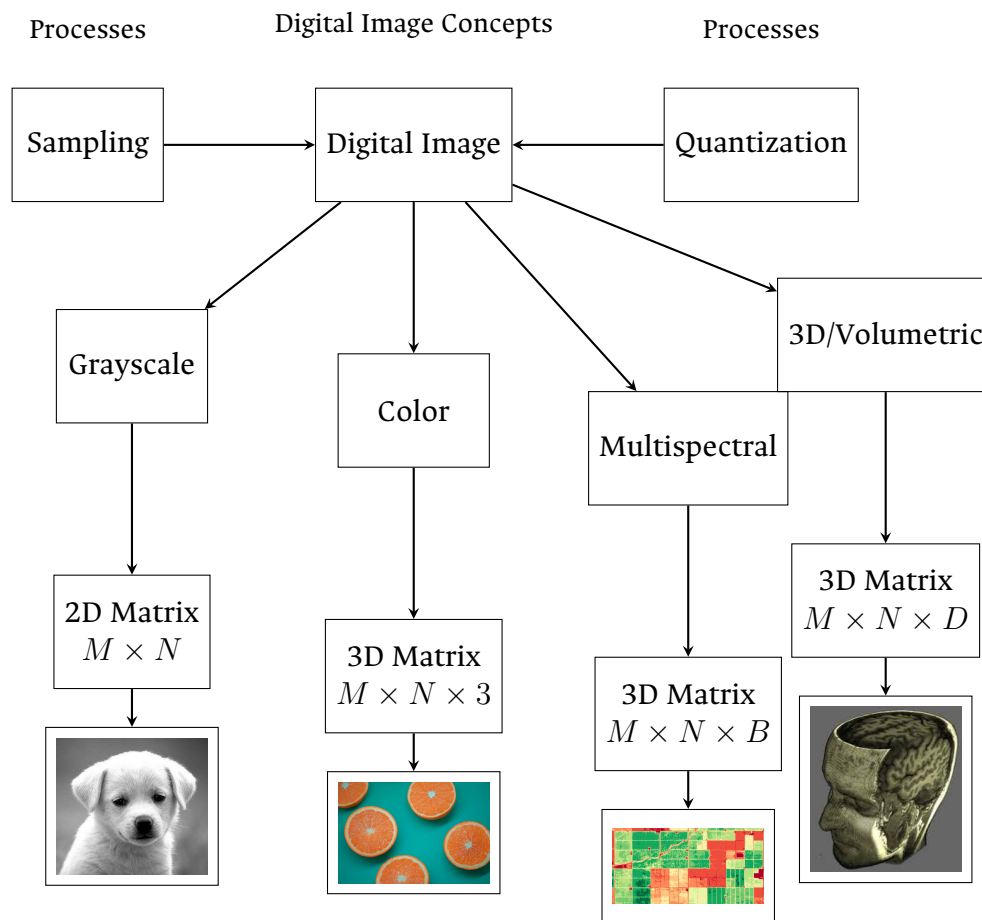


Figure 2-1 Overview of digital image concepts and their mathematical representations. The figure shows the main types of digital images (grayscale, color, multispectral, and volumetric), their mathematical representations, and the fundamental processes of sampling and quantization. Example images are included to illustrate each type.

2.2 Digital Histopathological Images

Digital histopathology represents a significant advancement in medical imaging, where traditional glass slides containing tissue samples are digitized using specialized scanning devices. This transformation has revolutionized the field of pathology by enabling remote diagnosis, computer-aided analysis, and digital archiving of tissue samples [Amgad et al., 2019]. This process has evolved significantly over the past few decades, as shown in Figure 2-2.

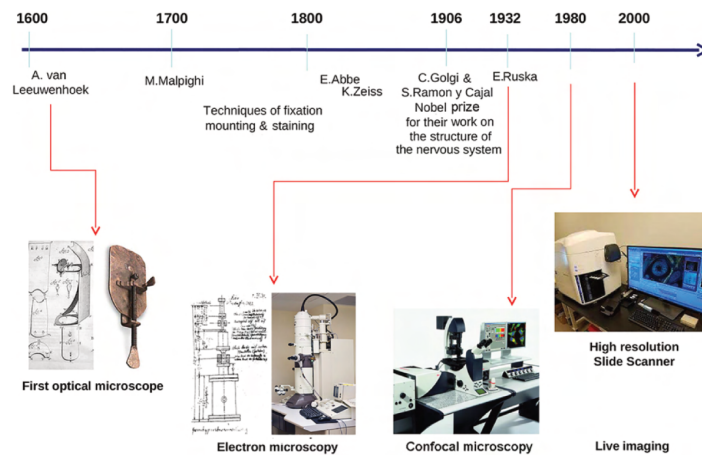


Figure 2-2 Histology evolution timeline. (Image from [Mazzarini et al., 2021]).

2.2.1 Whole Slide Imaging (WSI)

Whole Slide Imaging (WSI) is the process of digitizing entire glass slides at high resolution, creating a digital representation that can be viewed, analyzed, and shared electronically. Modern WSI scanners use sophisticated optical systems that capture multiple fields of view at high magnification, which are then stitched together to create a seamless digital image [Hu et al., 2025]. These systems incorporate high-resolution objectives with magnifications ranging from 20x to 40x, precise motorized stages for accurate slide positioning, automated focus

841 systems to maintain image quality, and high-quality cameras equipped with large
842 sensor arrays. The resulting digital slides can reach sizes of several gigabytes,
843 containing billions of pixels that capture the microscopic details of tissue samples
844 [Hu et al., 2025]. Figure 2-3 shows a whole slide imaging system by Omnyx for
845 slide digitization and a comprehensive digital pathology interface from Omnyx
846 designed to streamline pathologists' diagnostic workflow [Farahani et al., 2015].

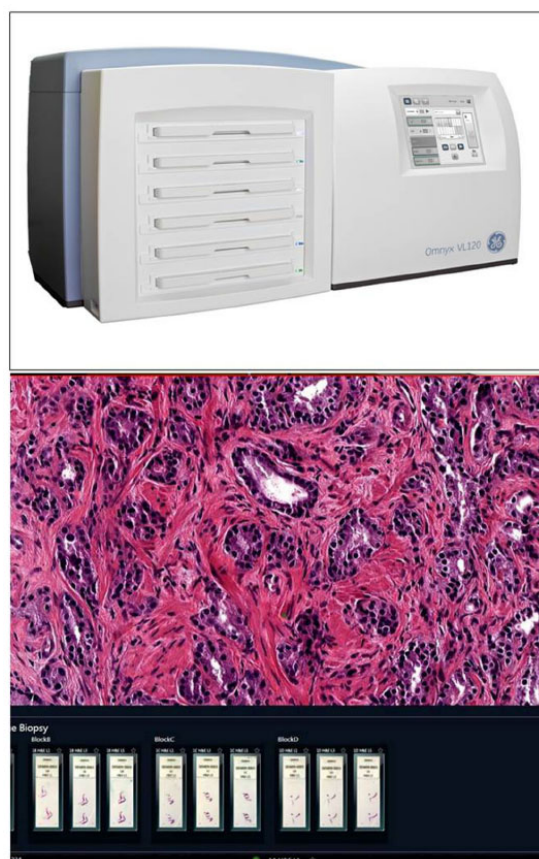


Figure 2-3 (Above) Whole slide imaging system by Omnyx for slide digitization. (Below) Comprehensive digital pathology interface from Omnyx designed to streamline pathologists' diagnostic workflow. (From [Farahani et al., 2015]).

847 2.2.2 Regions of Interest (ROI)

848 In digital histopathology, **Region of Interests (ROIs)** are specific areas within a
849 whole slide image that contain diagnostically relevant information. These regions
850 can be manually annotated by pathologists, automatically detected using
851 computer vision algorithms, or defined based on specific tissue characteristics or
852 abnormalities. The importance of **ROIs** lies in their ability to focus computational
853 analysis on relevant areas, reduce computational complexity in automated
854 systems, facilitate targeted diagnosis and research, and enable efficient storage and
855 transmission of critical information.

856 2.2.3 Staining Techniques

857 Histopathological analysis relies heavily on various staining techniques to enhance
858 the visibility of different tissue components and cellular structures. The choice of
859 staining method depends on the specific diagnostic requirements and the type of
860 tissue being examined.

861 Hematoxylin and Eosin (H&E)

862 Hematoxylin and Eosin (H&E) staining is the most widely used technique in
863 histopathology, particularly in breast cancer diagnosis [Pan et al., 2021]. This
864 staining method provides essential visualization through two components:
865 hematoxylin, which stains cell nuclei blue/purple to highlight nuclear morphology,
866 and eosin, which stains cytoplasm and extracellular matrix pink/red to reveal
867 tissue architecture.

868 The popularity of H&E staining in breast cancer histopathology stems from its
869 ability to clearly visualize tumor architecture and growth patterns, distinguish
870 between different types of breast cancer, identify important diagnostic features

871 like nuclear pleomorphism, and assess tumor grade and stage. Beyond breast
872 cancer, H&E staining finds extensive application across various medical specialties
873 including general pathology, dermatology, gastroenterology, neurology, and
874 oncology.

875 Special Stains

876 In addition to H&E, various special stains are used for specific diagnostic purposes.
877 Immunohistochemistry (IHC) uses antibodies to detect specific proteins, playing a
878 crucial role in subtyping breast cancers. Key IHC stains include Estrogen Receptor
879 (ER) staining for detecting estrogen receptors, Progesterone Receptor (PGR)
880 staining for assessing progesterone receptor status, Human Epidermal Growth
881 Factor Receptor 2 (HER2) staining for evaluating HER2 protein expression, and
882 Ki67 staining for measuring cellular proliferation rates. These markers are
883 particularly crucial in breast cancer diagnosis and treatment planning, as they help
884 determine the molecular subtype of the cancer and guide personalized therapeutic
885 approaches. Other specialized stains include Periodic Acid-Schiff (PAS) for
886 highlighting carbohydrates and basement membranes, Masson's Trichrome for
887 distinguishing between collagen and muscle fibers, and silver stains for detecting
888 microorganisms and nerve fibers. These specialized staining techniques
889 complement H&E by providing additional diagnostic information that is crucial for
890 accurate diagnosis and treatment planning [Weitz et al., 2023]. Examples of these
891 staining techniques are shown in Figure 2-7.

892 2.3 Deep Learning Fundamentals

893 Deep Learning has emerged as a powerful subset of machine learning,
894 revolutionizing the field of artificial intelligence. Its roots can be traced back to the
895 early development of artificial neural networks in the 1940s and 1950s, with
896 significant milestones including the perceptron in 1958 and the backpropagation

algorithm in the 1980s. However, it wasn't until the early 21st century, with the advent of more powerful computational resources and the availability of large datasets, that deep learning truly began to flourish.

2.3.1 Learning Paradigms

Deep Learning systems can be categorized into three main learning paradigms. The most common approach is supervised learning, where models learn from labeled data by mapping inputs to known outputs. This paradigm requires a large amount of labeled training data, which can be expensive and time-consuming to acquire. Semi-supervised learning offers a hybrid approach that leverages both labeled and unlabeled data, proving particularly useful when labeled data is scarce but unlabeled data is abundant. Finally, unsupervised learning enables models to discover patterns and structures from unlabeled data without explicit guidance, making it valuable for tasks like clustering and dimensionality reduction.

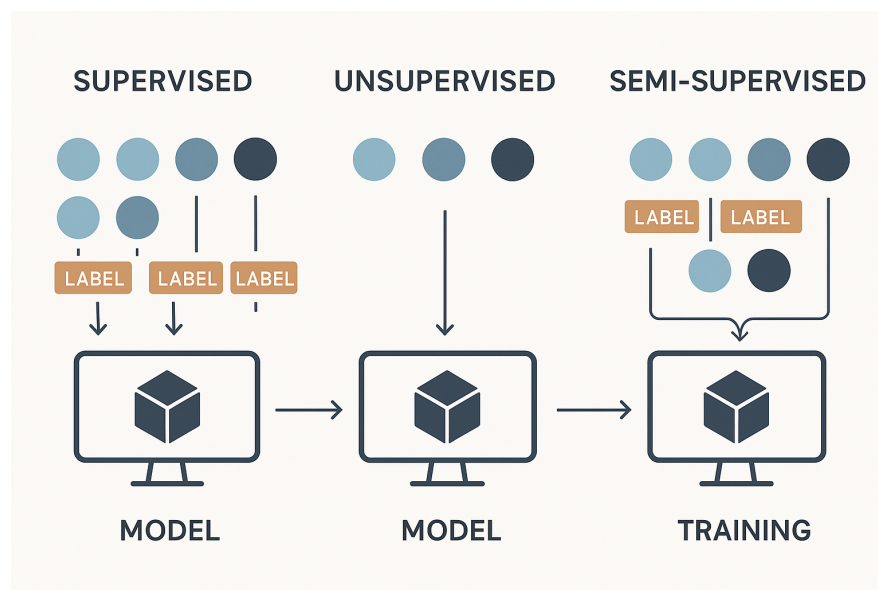


Figure 2-4 Common learning paradigms.

910 2.3.2 Architecture and Training

911 Deep Learning architectures are characterized by their layered structure, where
912 each layer progressively extracts and transforms features from the input data. The
913 early layers typically focus on low-level feature extraction, such as edges, textures,
914 and basic patterns in the case of image processing. As information flows through
915 the network, middle layers combine these basic features into more complex
916 representations. The final layers perform high-level reasoning and make the
917 ultimate predictions or classifications.

918 The training process relies heavily on the gradient descent algorithm, which
919 iteratively adjusts the model's parameters to minimize a loss function. This loss
920 function serves as a crucial component of the learning process, quantifying how
921 well the model's predictions match the actual targets. By providing a measure of
922 the model's performance, the loss function guides the optimization process,
923 enabling the network to learn meaningful patterns from the training data.

924 2.3.3 Challenges and Solutions

925 Despite their power, deep learning systems face several significant challenges. One
926 of the most prominent issues is overfitting, where models may memorize training
927 data instead of learning generalizable patterns. This challenge is typically
928 addressed through various regularization techniques such as dropout, L1/L2
929 regularization, and early stopping. Another critical challenge is the substantial
930 data requirements; deep learning models often need massive amounts of training
931 data to achieve good performance, which can be a limiting factor in many
932 applications. Additionally, the complex, layered nature of deep learning models
933 makes them difficult to interpret, often referred to as "black boxes." This lack of
934 transparency can be particularly problematic in critical applications where
935 understanding the decision-making process is essential.

936 2.3.4 Deep Learning Frameworks

937 The development of powerful open-source frameworks has significantly
938 accelerated deep learning research and applications. TensorFlow, developed by
939 Google, provides a comprehensive ecosystem for building and deploying machine
940 learning models [Abadi et al., 2016]. PyTorch, created by Facebook’s AI Research
941 lab, offers dynamic computation graphs and has become particularly popular in
942 research settings. Caffe, known for its speed and modularity, is widely used in
943 computer vision applications.

944 These frameworks have democratized deep learning by providing efficient
945 implementations of common operations, automatic differentiation for gradient
946 computation, and GPU acceleration for faster training. They also offer pre-trained
947 models and transfer learning capabilities, along with active communities for
948 support and knowledge sharing. The combination of these frameworks with
949 modern hardware has enabled researchers and practitioners to develop
950 increasingly sophisticated models, pushing the boundaries of what’s possible in
951 artificial intelligence. As shown in Figure 2-5, which presents data from Google
952 Trends over the last five years (as of April 2025), TensorFlow and PyTorch have
953 emerged as the two most prominent frameworks in the deep learning landscape.

954 2.4 Datasets and Data Sources

955 Throughout the development of this work, multiple datasets were used for
956 evaluation of ISS models. The common elements of all these datasets are that they
957 contain RGB images and are crowdsourced with multiple labelers, where not
958 necessarily all labeler label all images.

959 As it has been mentioned in Chapter 1, the main goal of this work is mainly
960 focused on crowdsourced histopathology images semantic segmentation, however,
961 these datasets present the following challenges:

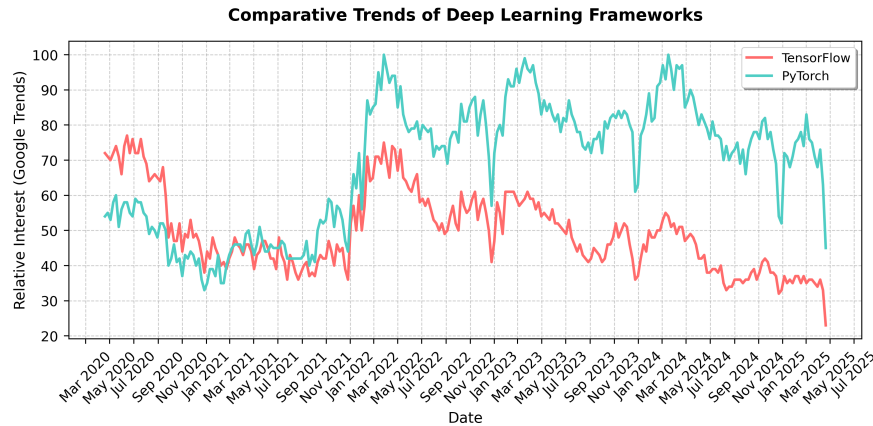


Figure 2-5 Comparative Trends of the top two most popular Deep Learning Frameworks, apparently, tendency was switched to PyTorch since 2022

- Distribution of segmentation labels is not uniform across the image, since some tissues and structures are more common than others.
- Visualization of performance of the models in debug time (like per epoch analysis) is not simple for non experts in the subject, which makes it hard to evaluate whether the model is overfitting or not at a glance.

For these reasons, multiple datasets were created in the pursuit of an initial evaluation of performance of the models against more traditional and familiar images before the focus on histopathology images. Once a decent performance in metrics like Dice coefficient was achieved, the focus was shifted to histopathology images and further tunings on the models were performed if needed.

In any case, both the emulated noisy annotations datasets and the histopathology datasets somehow contained ground truth aggregation, either from the original source (in the case of emulated noisy annotations), the expert annotation (if available) or from the aggregation of multiple labelers ¹.

¹STAPLE in the case of histopathology datasets with no expert annotations available

2.4.1 Oxford-IIIT Pet Dataset

Using the techniques described later in Section 5.3.1, the Oxford-IIIT Pet Dataset [Parkhi et al., 2012] was used to create a dataset with emulated noisy annotations. This dataset is officially available in the ‘tensorflow_datasets’ package, making it convenient to use in experiments. It is composed of 7349 images of pets, being 2371 of these images cats, and the rest 4978, dogs. Breed labels are included as well as segmentation masks for the classes: ‘background’, ‘border’, and ‘pet’. Figure 2-6 shows an example of the annotations in the original dataset.

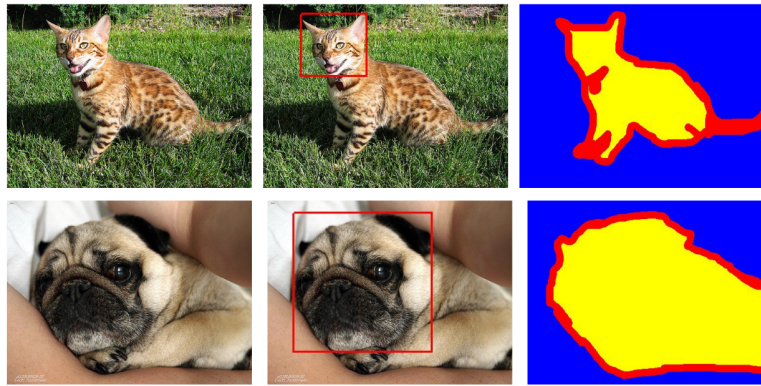


Figure 2-6 Annotations in the Oxford-IIIT Pet data. From left to right: pet image, head bounding box, and trimap segmentation (blue: background region; red: ambiguous region; yellow: foreground region).

2.4.2 Real histopathology datasets

Multi-Stain Breast Cancer Histological Dataset

The Multi-Stain Breast Cancer Histological Dataset [Weitz et al., 2023] represents one of the largest publicly available collections of whole slide images (WSIs) from surgical resection specimens of primary breast cancer patients. This dataset is particularly valuable for our work because it contains matched pairs of H&E and

IHC-stained tissue sections from the same tumor, with a total of 4,212 WSIs from 1,153 patients. The IHC stains include important biomarkers such as ER, PGR, HER2, and KI67, which are routinely used in breast cancer diagnosis and treatment planning (more on staining techniques in Section 2.2.3).

The dataset’s relevance to our work stems from several key aspects. The matched H&E and IHC stains allow for studying the consistency of segmentation across different staining modalities, which is crucial for understanding how different visualization methods affect annotation quality. With 1,153 patients, the dataset provides a robust foundation for training and evaluating segmentation models in a real-world clinical setting. The inclusion of routine diagnostic cases makes the dataset representative of actual clinical practice, where variations in staining quality and tissue preparation are common. Furthermore, the multiple biomarker stains (ER, PGR, HER2, KI67) enable the study of how different tissue characteristics affect segmentation performance and annotator agreement.

This dataset serves as an ideal testbed for our crowdsourced segmentation approach. It allows us to evaluate how different staining modalities affect annotator performance and agreement, while also providing insights into the relationship between tissue characteristics and segmentation difficulty. The dataset’s comprehensive nature enables validation of our models’ performance across different biomarker expressions and assessment of the generalizability of our approach to real-world clinical data.

Structured Crowdsourcing Dataset for Histology Images

The dataset presented by [Amgad et al., 2019] is particularly relevant to our work as it represents one of the first systematic studies of crowdsourced annotations in histopathology. The authors recruited 25 participants with varying levels of expertise (from senior pathologists to medical students) to delineate tissue regions in 151 breast cancer slides using the Digital Slide Archive platform, resulting in over 20,000 annotated tissue regions.

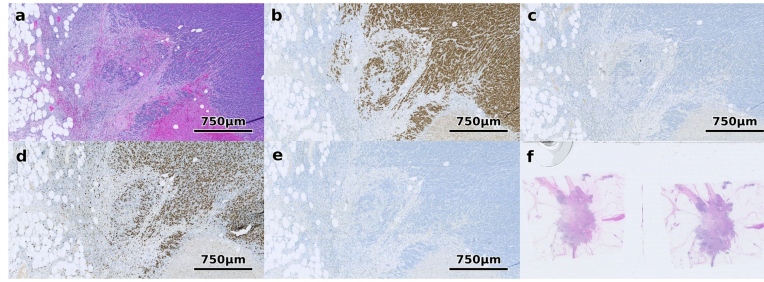


Figure 2-7 Different staining techniques obtained from multi-stain breast cancer dataset [Weitz et al., 2023]. (a) shows H&E, (b) ER, (c) HER2, (d) Ki67 and (e) PGR. (f) shows an example of a WSI that was excluded since it contains multiple tissue sections.

Key aspects of this dataset make it valuable for our work. The systematic evaluation of inter-participant discordance revealed varying levels of agreement across different tissue classes, with low discordance for tumor and stroma, and higher discordance for more subjectively defined or rare tissue classes. The inclusion of feedback from senior participants helped in curating high-quality annotations, demonstrating that fully convolutional networks trained on these crowdsourced annotations can achieve high accuracy (mean AUC=0.945). The dataset also provides evidence that the scale of annotation data significantly improves image classification accuracy.

This dataset is particularly valuable for our work because it provides crucial insights into how annotator expertise affects segmentation quality. It demonstrates the feasibility of using crowdsourced annotations for training accurate segmentation models, showing that even with varying levels of expertise, aggregated annotations can produce reliable ground truth. The dataset includes a systematic analysis of inter-annotator agreement, which is crucial for understanding the challenges in crowdsourced histopathology segmentation and informing the development of more robust segmentation approaches.

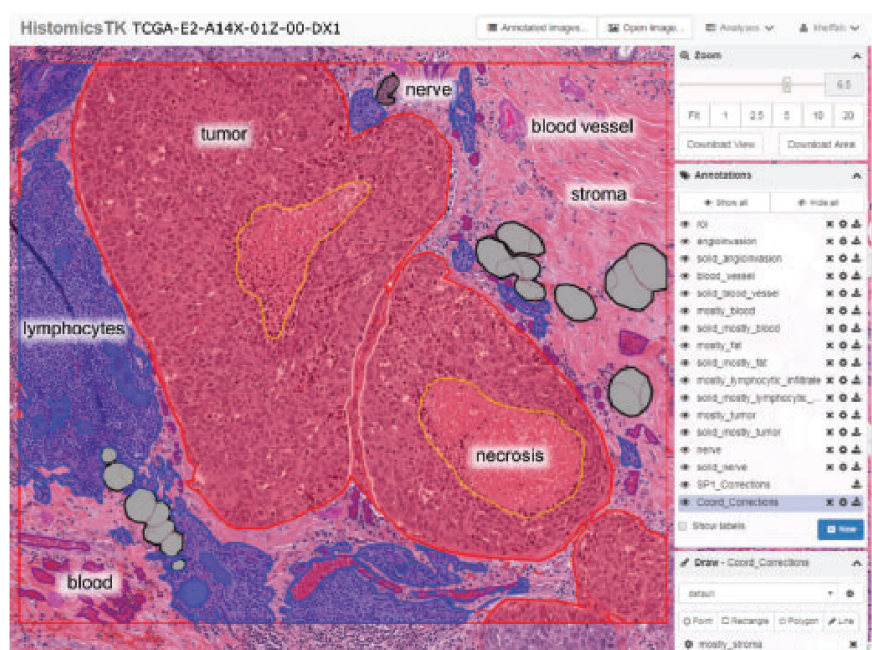


Figure 2-8 Screenshot of the DSA and HistomicsTK web interface while creating the crowdsourced annotations for the dataset presented by [Amgad et al., 2019].

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CHAPTER

THREE

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CHAINED GAUSSIAN PROCESSES

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3.1 Background and Related Methods

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In recent years, several approaches have been proposed to handle supervised learning problems in the context of multiple annotators. This section reviews key methods that form the foundation for understanding chained Gaussian processes and their application to multiple annotator scenarios in regression and classification tasks.

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3.1.1 Kernel Alignment-Based Annotator Relevance Analysis (KAAR)

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The Kernel Alignment-Based Annotator Relevance Analysis (KAAR) method, introduced in [Julián and Álvarez Meza Andrés Marino, 2023], addresses the challenge of estimating annotator expertise in scenarios where the ground truth is

1051 unavailable. The key innovation of KAAR lies in its use of Centered Kernel
 1052 Alignment (CKA) to measure the similarity between input features and annotator
 1053 labels.

1054 Given a dataset with input features $\mathbf{X}$ and labels from multiple annotators $\mathbf{Y}$, KAAR
 1055 computes a kernel matrix $\mathbf{K}_\nu$ as a convex combination of R basis kernels:

$$\mathbf{K}_\nu = \sum_{r=1}^R \nu_r \mathbf{K}_r, \quad (3-1)$$

1056 where $\mathbf{K}_r$ is computed using a kernel function over the input features, and ν_r
 1057 represents the relevance weight for the r -th annotator. The weights are optimized
 1058 by maximizing the CKA between $\mathbf{K}_\nu$ and a target kernel $\mathbf{F}$ computed from the
 1059 input features:

$$\rho(\mathbf{K}_\nu, \mathbf{F}) = \frac{\langle \bar{\mathbf{K}}_\nu, \bar{\mathbf{F}} \rangle_{\text{F}}}{\| \bar{\mathbf{K}}_\nu \|_{\text{F}} \| \bar{\mathbf{F}} \|_{\text{F}}}, \quad (3-2)$$

1060 A key advantage of KAAR is its ability to capture dependencies between annotators
 1061 through the kernel framework, without requiring explicit parametric modeling of
 1062 these relationships.

1063 3.1.2 Localized Kernel Alignment-Based Annotator 1064 Relevance Analysis (LKAAR)

1065 LKAAR extends KAAR by introducing locality in the kernel alignment framework.
 1066 Rather than assuming constant annotator performance across the input space,

1067 LKAAR models the annotator relevance as a function of the input features. The
 1068 kernel combination in LKAAR takes the form:

$$K_q = \sum_{r=1}^R Q_r K_r Q_r, \quad (3-3)$$

1069 where Q_r is a diagonal matrix whose elements represent the local relevance of
 1070 annotator r . These elements are computed as:

$$q_r(\mathbf{x}_n) = \begin{cases} \beta_0^{(r)} + \sum_{n'=1}^N \beta_{n'}^{(r)} \kappa_\beta(\mathbf{x}_n, \mathbf{x}_{n'}), & \text{if } n \in N_r \\ 0, & \text{Otherwise} \end{cases} \quad (3-4)$$

1071 This localized approach allows LKAAR to better model inconsistent annotators
 1072 whose performance varies across different regions of the input space.

1073 **3.1.3 Regularized Chained Deep Neural Network** 1074 **(RCDNN)**

1075 RCDNN represents a deep learning approach to multiple annotator learning,
 1076 inspired by the chained Gaussian processes model. The method models both the
 1077 ground truth estimation and annotator reliability through a deep neural network
 1078 architecture. The likelihood function in RCDNN takes the form:

$$p(\mathbf{Y}|\boldsymbol{\theta}) = \prod_{n=1}^N \prod_{r \in R_n} \left(\prod_{k=1}^K \zeta_{n,k}^{\delta(y_n^{(r)} - k)} \right)^{\lambda_n^{(r)}} \left(\frac{1}{K} \right)^{1 - \lambda_n^{(r)}}, \quad (3-5)$$

1079 where $\lambda_n^{(r)}$ represents the reliability of annotator r for instance n , and $\zeta_{n,k}$ is the
 1080 estimated probability of class k for instance n .

1081 The RCDNN architecture is built upon a sophisticated multi-layer structure that
 1082 combines several key innovations in deep learning. At its core, it employs multiple
 1083 dense layers with carefully chosen activation functions, each designed to capture
 1084 different aspects of the annotator-data relationship. To prevent overfitting, a
 1085 comprehensive regularization strategy is implemented, combining both l1 and l2
 1086 norms with dropout mechanisms. The network architecture splits into specialized
 1087 branches: one dedicated to estimating the ground truth labels, and another
 1088 focused on modeling annotator reliability. This dual-branch approach allows the
 1089 network to simultaneously learn both the true underlying patterns in the data and
 1090 the individual characteristics of each annotator. Furthermore, the implementation
 1091 of Monte Carlo dropout during prediction provides a mechanism for uncertainty
 1092 estimation, offering not just predictions but also confidence measures for those
 1093 predictions.

1094 3.1.4 Chained Gaussian Processes Model

1095 The Chained Gaussian Processes (CGP) model provides a probabilistic framework for
 1096 modeling multiple outputs through a chain of Gaussian processes. In the context of
 1097 multiple annotators, CGP can be used to model both the ground truth and annotator
 1098 behavior. The key idea is to model a likelihood function with J parameters as in
 1099 [Giraldo et al., 2022]:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \prod_{n=1}^N p(y_n | \theta_1(\mathbf{x}_n), \dots, \theta_J(\mathbf{x}_n)), \quad (3-6)$$

1100 where each parameter $\theta_j(\mathbf{x})$ is modeled using a separate Gaussian process. This
 1101 framework provides remarkable flexibility in modeling complex relationships

1102 within the data. The use of Gaussian processes enables natural uncertainty
1103 quantification through posterior distributions, allowing the model to express
1104 varying degrees of confidence in its predictions. The framework inherently
1105 handles missing data through its probabilistic nature, and the automatic relevance
1106 determination property of GPs helps identify the most important features for
1107 prediction.

1108 The CGP framework becomes particularly powerful when integrated with concepts
1109 from KAAR and LKAAR. By incorporating kernel alignment principles, the model
1110 can make more informed choices about kernel functions, better capturing the
1111 underlying structure of the data. The framework can be extended to include
1112 input-dependent lengthscales, allowing for varying degrees of smoothness across
1113 the input space. Furthermore, through multi-output GP formulations, the model
1114 can capture complex dependencies between different annotators, providing a rich
1115 representation of how different experts' opinions relate to each other.

1116 However, implementing CGP in practice presents several significant challenges.
1117 The computational complexity of the model scales cubically with the dataset size,
1118 making it potentially prohibitive for large-scale applications. The need for
1119 approximate inference techniques in non-Gaussian likelihood scenarios can
1120 introduce additional complexity and potential approximation errors. When
1121 dealing with high-dimensional inputs, the model may struggle with the curse of
1122 dimensionality, requiring careful feature selection or dimensionality reduction
1123 preprocessing. Additionally, the selection of appropriate kernel functions remains
1124 a critical challenge, as it significantly impacts the model's performance and its
1125 ability to capture relevant patterns in the data.

1126 3.2 Tooling and Implementation

1127 The implementation of Chained Gaussian Processes for multiple annotator
1128 learning requires robust and flexible software tools that can handle both the

1129 probabilistic nature of Gaussian processes and the complexity of deep
1130 architectures. In our implementation, we leverage two key frameworks: GPflow
1131 and GPflux. These tools provide the foundation for building scalable and efficient
1132 implementations of our models while maintaining the theoretical rigor necessary
1133 for proper uncertainty quantification.

1134 The choice of these frameworks is motivated by several factors. First, both are
1135 built on top of TensorFlow, providing access to automatic differentiation and GPU
1136 acceleration capabilities. Second, they offer a modular design that allows for easy
1137 extension and modification, which is crucial when implementing novel model
1138 architectures. Third, they provide robust implementations of various Gaussian
1139 process models and inference methods, allowing us to focus on the specific
1140 challenges of multiple annotator learning rather than reimplementing basic GP
1141 functionality.

1142 3.2.1 GPflow Overview

1143 GPflow is a Python package for building Gaussian process models in TensorFlow.
1144 Developed as a modern and scalable alternative to GPy, it leverages TensorFlow’s
1145 computational capabilities to provide efficient implementation of Gaussian process
1146 models. The framework is designed with four key principles in mind: ease of use,
1147 computational efficiency, extensibility, and automatic differentiation. This work
1148 was originally published in [de G. Matthews et al., 2016].

1149 At its core, GPflow provides implementations of various Gaussian process models
1150 including GPR (Gaussian Process Regression), GPMC (Gaussian Process Monte
1151 Carlo), SGPR (Sparse Gaussian Process Regression), and SGPMC (Sparse Gaussian
1152 Process Monte Carlo). These implementations are built on top of TensorFlow’s
1153 computational graph framework, allowing for automatic differentiation and GPU
1154 acceleration where available.

The framework’s architecture is modular, with clear separation between model components such as kernels, likelihoods, and optimization routines. This modularity makes it straightforward to extend existing models or create new ones. GPflow includes a comprehensive suite of kernel functions, ranging from the basic RBF (Radial Basis Function) kernel to more sophisticated ones like the Spectral Mixture kernel.

One of GPflow’s distinguishing features is its robust handling of model parameters. It provides facilities for parameter transformation (ensuring parameters remain in valid ranges during optimization), prior specification, and flexible optimization strategies. The framework also includes utilities for model evaluation and prediction, with built-in support for uncertainty estimation.

3.2.2 GPflux Framework

GPflux extends GPflow to enable the construction and training of Deep Gaussian Process (DGP) models using modern deep learning practices. Built on top of both GPflow and TensorFlow, GPflux provides a flexible framework for implementing deep probabilistic models that combine the expressiveness of deep neural networks with the uncertainty quantification capabilities of Gaussian processes. This work was originally published in [Dutordoir et al., 2021].

The framework implements Deep Gaussian Processes as a composition of Gaussian process layers, where each layer transforms its inputs through a Gaussian process mapping. This hierarchical structure allows the model to capture complex, non-stationary patterns in data while maintaining uncertainty estimates throughout the entire processing pipeline. GPflux handles this complexity through a carefully designed architecture that separates the concerns of model specification, inference, and prediction.

A key innovation in GPflux is its integration with Keras-style layers, allowing practitioners to combine deterministic neural network layers with Gaussian

process layers in a single model. This hybrid approach enables the construction of models that can leverage both the structural assumptions encoded in neural network architectures and the flexibility of Gaussian processes.

For inference, GPflux implements various approximate methods, including stochastic variational inference with inducing points, which makes it possible to train deep Gaussian process models on large datasets. The framework provides built-in support for minibatch training, making it practical to work with datasets that don't fit in memory.

3.3 Experimental Setup

Synthetic datasets were used to evaluate the proposed methods in classification and regression tasks, following very similar procedures to the ones used in [Gil-Gonzalez et al., 2023], which are as follows:

3.3.1 Classification

A fully synthetic data for a 1-D ($P = 1$) multiclass classification problem with $K = 3$ classes is generated. The input feature matrix $\mathbf{X}$ is constructed by randomly sampling $N = 100$ points from a uniform distribution over the interval $[0, 1]$. For each sample n , the true label is determined by computing three target values:

$$\begin{aligned} t_{n,1} &= \sin(2\pi x_n), \\ t_{n,2} &= -\sin(2\pi x_n), \\ t_{n,3} &= -\sin(2\pi(x_n + 0.25)) + 0.5, \end{aligned} \tag{3-7}$$

and assigning the class label as the index $i \in \{1, 2, 3\}$ that maximizes $t_{n,i}$. This construction ensures a nontrivial, nonlinear decision boundary between the classes.

For evaluation, the test set is generated by extracting 200 equally spaced samples from the interval $[0, 1]$, providing a dense coverage of the input space to assess generalization performance.

Note that at this point, the fully synthetic dataset does not contain real annotator labels. Therefore, it is necessary to simulate annotator labels as corrupted versions of the hidden ground truth. In these experiments, the simulation is designed to account for: 1) dependencies among annotators, and 2) labeler performance that varies as a function of the input features.

To achieve this, an **SLFM**-based approach (termed SLFM-C) is employed, which generates annotator labels as follows:

1. **Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q : \mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R$, $n \in N$.
2. **Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$, where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D representation of the input features in $\mathbf{X}$, obtained using the t-distributed Stochastic Neighbor Embedding (t-SNE) approach.
3. **Reliability calculation:** Calculate $\hat{\lambda}_n^r = \varsigma(\hat{f}_{l,r,n})$, where $\varsigma(\cdot) \in [0, 1]$ is the sigmoid function.
4. **Label assignment:** The r th annotator's label for the n th sample is given by

$$y_n^r = \begin{cases} y_n, & \text{if } \hat{\lambda}_n^r \geq 0.5 \\ \tilde{y}_n, & \text{if } \hat{\lambda}_n^r < 0.5 \end{cases}$$

where y_n is the true label and $\tilde{y}_n$ is a flipped version of the actual label y_n .

This procedure ensures that the simulated annotator labels exhibit both inter-annotator dependencies and input-dependent reliability.

1223 3.3.2 Regression

1224 A fully synthetic dataset is also generated for a 1-D regression problem, where the
1225 ground truth for the n th sample is defined as:

$$y_n = \sin(2\pi x_n) \sin(6\pi x_n) \quad (3-8)$$

1226 Here, the input matrix $\mathbf{X}$ is constructed by randomly sampling 100 points uniformly
1227 from the interval $[0, 1]$. The test set is created by extracting equally spaced samples
1228 from the same interval, ensuring a clear separation between training and testing
1229 data.

1230 To simulate the presence of multiple annotators, the ground truth is corrupted with
1231 annotator-specific noise. For each annotator r , the observed label for the n th sample
1232 is given by:

$$y_n^r = y_n + \epsilon_n^r$$

1233 where $\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$ is Gaussian noise, and v_n^r represents the error variance for
1234 the r th annotator, which may depend on the input features. This setup allows us
1235 to model both the bias and the reliability of each annotator, as well as potential
1236 dependencies between annotators.

1237 To further increase realism, the error variance v_n^r can be modeled as a function of
1238 the input features, capturing scenarios where annotator reliability varies across the
1239 input space, thus:

$$\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$$

1240 being v_n^r , the r th annotator's error variance for the n th sample.

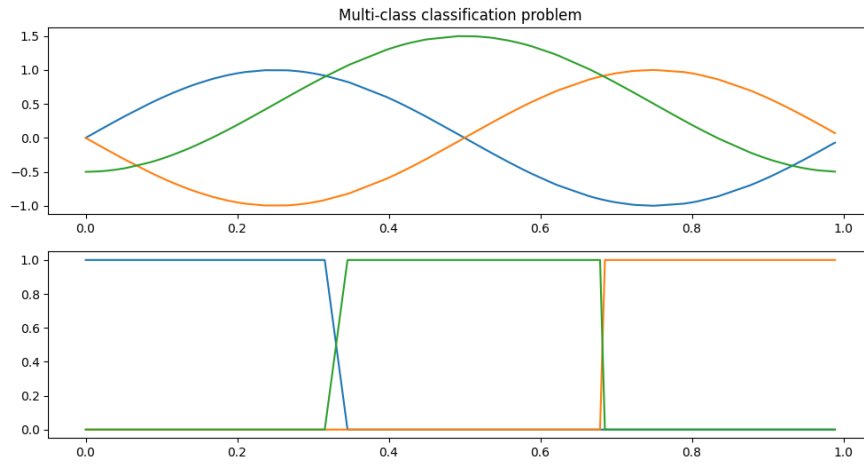
1241 Again, for modeling the error as a function of the input features, an **SLFM**-based
1242 approach (termed SLFM-R) is employed to build the noisy labels, as follows:

- 1243 1. **Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q :$
1244 $\mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R, n \in N$.
- 1245 2. **Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$,
1246 where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D
1247 representation of the input features in $\mathbf{X}$, obtained using the t-distributed
1248 Stochastic Neighbor Embedding (t-SNE) approach.
- 1249 3. **Compute variance :** Finally, determine $v_n^r = \exp(\hat{f}_{l,r,n})$.

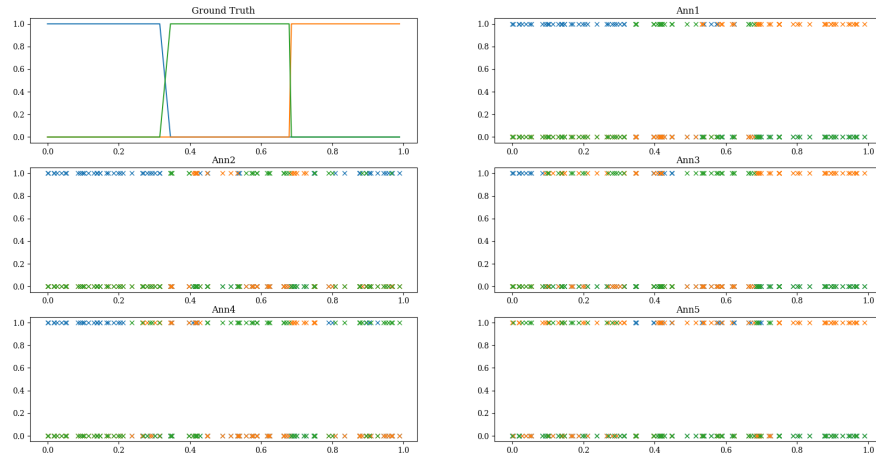
1250 3.4 Results

1251 The built synthetic datasets are used to train the **CCGPMA** model. Emulated
1252 classification labels following the true label determining function from 3-7 can be
1253 seen in Figure 3-1. On the other hand, the ground truth and noisy labels emulation
1254 for annotators across the feature space for regression task can be seen in Figure
1255 3-3.

1256 Using both GPflux and GPflow for the regression and classifications tasks, leads to
1257 the following combinations experiments: GPflux-CCGPMA Regression,
1258 GPflow-CCGPMA Regression, GPflux-CCGPMA Binary Classification,
1259 GPflow-CCGPMA Binary Classification, GPflow-CCGPMA Multiclass Classification,
1260 GPflux-CCGPMA Multiclass Classification. All of the experiments converged
1261 successfully with almost identical results and close to no difference in
1262 performance.



(a) Ground truth labels across the feature space



(b) Noisy emulations for annotators across the feature space for classification task

Figure 3-1 Synthetic dataset overview for classification task

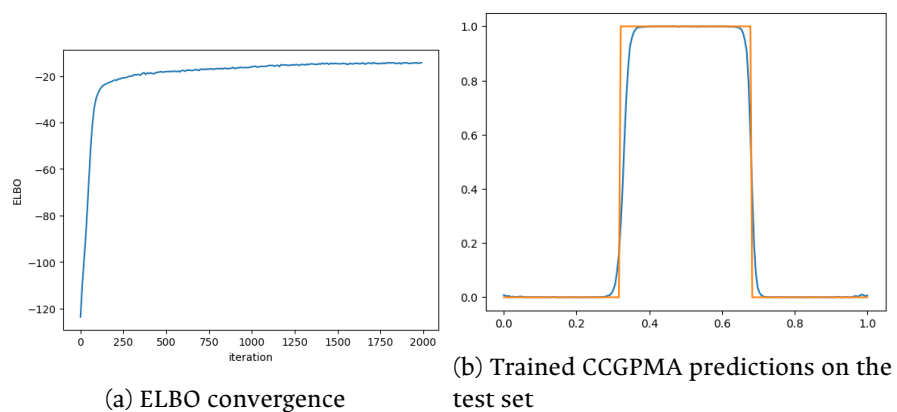


Figure 3-2 Classification results for the CCGPMA model using GPflux.

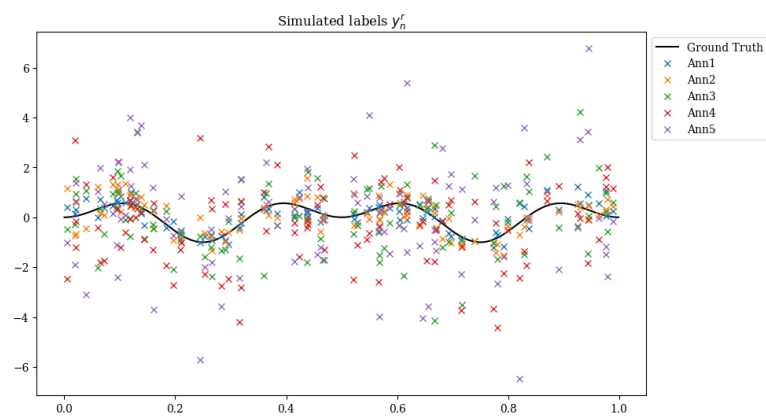
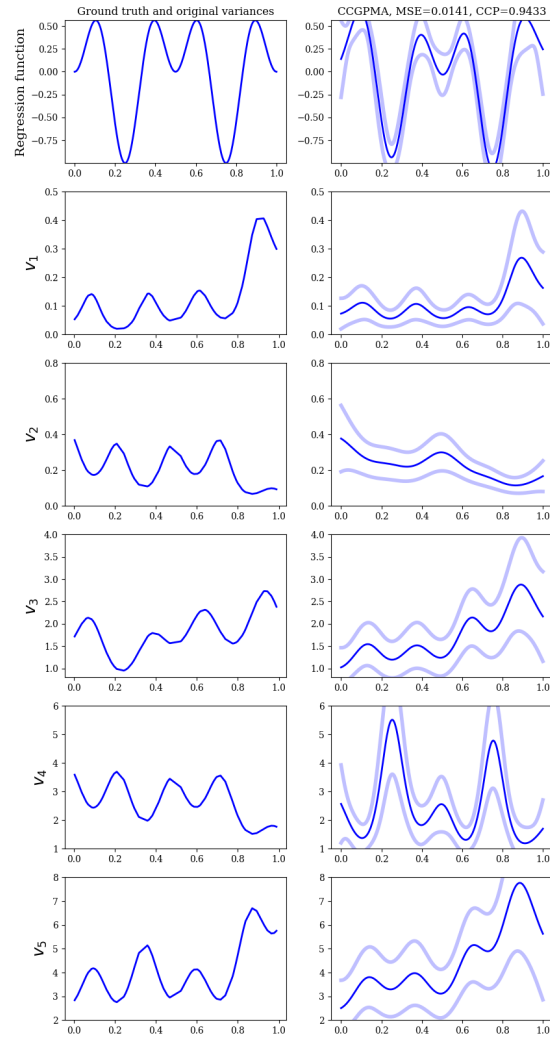


Figure 3-3 Ground truth and noisy labels emulation for annotators across the feature space for regression task.



(a) Trained CCGPMA predictions on the test set, for inference of both the hidden ground truth and the annotators' reliabilities across the feature space.

Figure 3-4 Regression results for the CCGPMA model using GPflux.

3.5 Conclusion

3.5.1 Comparative Analysis

The evolution of multiple annotator learning methods represents a fascinating progression in how we approach the challenge of learning from diverse expert opinions. KAAR established a foundation with its non-parametric approach through kernel alignment, though it operates under the assumption of constant annotator performance across the input space. LKAAR advanced this concept by introducing input-dependent annotator behavior, recognizing that expert performance often varies across different types of inputs. RCDNN brought the power and flexibility of deep learning to the problem, offering excellent scalability but requiring careful attention to regularization and architecture design. CGP represents perhaps the most theoretically elegant approach, providing a principled probabilistic framework, though it faces significant computational challenges in practical applications.

The selection of an appropriate method for a given application requires careful consideration of several factors. The size and dimensionality of the dataset play a crucial role, as some methods scale better than others with increasing data volume. The need for uncertainty quantification may favor probabilistic approaches like CGP, while computational constraints might push towards more efficient methods like KAAR. The presence of input-dependent annotator behavior could make LKAAR or RCDNN more appropriate choices. Additionally, when interpretability is a key requirement, as in many medical applications, methods like KAAR and LKAAR might be preferred over the more complex deep learning approaches. Table 3-1 provides a comprehensive comparison of the key characteristics and capabilities of each method discussed.

Table 3-1 Comparison of Multiple Annotator Learning Methods

Characteristic	KAAR	LKAAR	RCDNN	CGP
Input-dependent annotator behavior	No	Yes	Yes	Yes
Annotator dependencies modeling	Yes	Yes	Yes	Yes
Uncertainty quantification	No	No	Partial	Yes
Computational scalability	High	Medium	High	Low
Missing data handling	Yes	Yes	Yes	Yes
Non-linear relationships	Yes	Yes	Yes	Yes
Training complexity	Low	Medium	High	High
Hyperparameter sensitivity	Low	Medium	High	Medium
Interpretability	High	Medium	Low	Medium
Memory requirements	Low	Medium	High	High

Note: Partial uncertainty quantification in RCDNN refers to the use of Monte Carlo dropout for approximate uncertainty estimation.

1288 3.5.2 Implementation Considerations

1289 When implementing Chained Gaussian Processes for multiple annotator learning
 1290 using these tools, several key considerations must be addressed. First, the model
 1291 architecture must be carefully designed to capture both the individual annotator
 1292 characteristics and their interdependencies. This requires extending the basic GP
 1293 layers provided by GPflux to handle multiple output streams - one for each
 1294 annotator's reliability model.

1295 Second, the inference procedure must be adapted to handle the unique challenges
 1296 of multiple annotator data, such as missing labels and varying levels of annotator
 1297 expertise across different regions of the input space. The variational inference
 1298 capabilities of GPflux provide a solid foundation for this, but careful attention
 1299 must be paid to the design of the variational approximation to ensure it captures
 1300 the relevant uncertainty in both the ground truth and annotator reliability
 1301 estimates.

1302 Third, kernel selection and design presents a critical challenge in implementing the
 1303 CGP model. The choice of kernel functions significantly impacts the model's ability

to capture the underlying structure of the data and the relationships between annotators. Several considerations must be taken into account:

- **Kernel Complexity:** The model requires multiple kernel functions to capture different aspects of the data - one for the ground truth and additional kernels for each annotator's reliability model. This multi-kernel setup increases the complexity of kernel selection and optimization.
- **Input-Dependent Behavior:** Since annotator performance often varies across the input space, the kernel functions must be flexible enough to capture this non-stationary behavior. This may require using composite kernels or kernels with input-dependent lengthscales.
- **Inter-Annotator Dependencies:** The kernel design must account for potential correlations between different annotators' behaviors. This can be achieved through careful design of the kernel structure in the SLFM framework, where the combination of basis kernels must be chosen to effectively model these dependencies.
- **Computational Efficiency:** The choice of kernel functions also impacts computational performance. While more expressive kernels may better capture complex patterns, they often come with increased computational cost. This trade-off must be carefully considered, especially when dealing with large datasets.

Finally, computational efficiency becomes a critical concern when scaling to large datasets with multiple annotators. The minibatch training capabilities of both frameworks help address this, but additional considerations such as appropriate inducing point selection and batch size tuning become important for achieving good performance in practice.

1329 3.5.3 Final Remarks

1330 In this chapter, we have presented a comprehensive overview of the Chained
1331 Gaussian Processes model and its application to multiple annotator learning. We
1332 have discussed the theoretical underpinnings of the model, its implementation
1333 using GPflow and GPflux, and its evaluation on synthetic datasets. It was shown in
1334 [Gil-Gonzalez et al., 2023] that CCGPMA outperforms other method in state of the
1335 art methods in terms of accuracy and uncertainty quantification (see 1.3.1).

1336 While Gaussian Processes have demonstrated remarkable success in regression
1337 and classification tasks, their application to image segmentation presents
1338 significant computational challenges. The cubic computational complexity of GPs
1339 ($\mathcal{O}(N^3)$), which scales with the number of data points, becomes particularly
1340 problematic when dealing with high-resolution images where each pixel can be
1341 considered a data point. This limitation makes traditional GP approaches
1342 impractical for large-scale image segmentation tasks, where the number of pixels
1343 can easily reach into the millions. Alternative approaches, such as deep
1344 learning-based segmentation methods, often provide more computationally
1345 efficient solutions for these high-dimensional problems, though at the cost of
1346 uncertainty quantification capabilities that GPs naturally provide.

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CHAPTER

FOUR

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TRUNCATED GENERALIZED CROSS ENTROPY FOR SEGMENTATION

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4.1 Loss Functions for Multiple Annotators

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As mentioned in Section 2.3.2, a loss function is a key element for defining the objective function of a deep learning model. The categorical cross-entropy loss is a common loss function for classification tasks. However, in the case of multiple annotators, the categorical cross-entropy loss is not able to handle the varying reliability of the annotators. In this section, we will propose a loss function that is able to handle multiple annotators' segmentation masks while accounting for their varying reliability across different regions of the image.

1361 4.1.1 Generalized Cross Entropy

1362 The Generalized Cross Entropy (GCE) loss function was first introduced by [Zhang
1363 and Sabuncu, 2018] as a robust alternative to the standard cross-entropy loss,
1364 particularly effective in handling noisy labels. Let us first consider the Cross
1365 Entropy (CE) and Mean Absolute Error (MAE) loss functions:

$$MAE(\mathbf{y}, f(\mathbf{x})) = \|\mathbf{y} - f(\mathbf{x})\|_1 \quad (4-1)$$

$$CE(\mathbf{y}, f(\mathbf{x})) = \sum_{k=1}^K y_k \log(f_k(\mathbf{x})) \quad (4-2)$$

1366 where $y_k \in \mathbf{y}$, $f_k(\mathbf{x}) \in f(\mathbf{x})$, and $\|\cdot\|_1$ stands for the l_1 -norm. Of note, $\mathbf{1}^\top \mathbf{y} =$
1367 $\mathbf{1}^\top f(\mathbf{x}) = 1$, $\mathbf{1} \in \{1\}^K$ being an all-ones vector. In addition, the MAE loss can be
1368 rewritten for softmax outputs, yielding:

$$MAE(\mathbf{y}, f(\mathbf{x})) = 2(1 - \mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x}))) \quad (4-3)$$

1369 where $\odot$ stands for the element-wise product.

1370 The CE loss function exhibits several distinct characteristics that make it
1371 particularly sensitive to noisy labels. It is unbounded from above and heavily
1372 penalizes confident but wrong predictions, which can lead to instability in the
1373 presence of label noise. In contrast, the MAE loss offers a more robust alternative
1374 with its bounded nature and symmetric properties in softmax-based
1375 representations. While the MAE loss is more resilient to noisy labels, it tends to

1376 train more slowly due to its equal weighting of all mistakes regardless of
1377 confidence.

1378 The GCE loss function, as defined by the authors in [Zhang and Sabuncu, 2018],
1379 provides a flexible framework that bridges these two approaches:

$$GCE(\mathbf{y}, f(\mathbf{x})) = 2 \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q}, \quad (4-4)$$

1380 with $q \in (0, 1]$. Remarkably, the limiting case for $q \rightarrow 0$ in GCE is equivalent to the
1381 CE expression, and when $q = 1$, it equals the MAE loss. In addition, the GCE holds
1382 the following gradient with regard to θ :

$$\frac{\partial GCE(\mathbf{y}, f(\mathbf{x}; \theta) | k)}{\partial \theta} = -f_k(\mathbf{x}; \theta)^{q-1} \nabla_\theta f_k(\mathbf{x}; \theta). \quad (4-5)$$

1383 The GCE loss combines the best of both worlds, offering robustness to label noise
1384 while maintaining the convexity property necessary for optimization. The
1385 truncation parameter q provides a mechanism to control the sensitivity to outliers,
1386 allowing for fine-tuning of the loss function's behavior based on the specific
1387 characteristics of the dataset.

1388 4.1.2 Extension to Multiple Annotators

1389 In the context of multiple annotators, we need to consider the varying reliability
1390 of each annotator across different regions of the image. Let's consider a k -class
1391 multiple annotators segmentation problem with the following data representation:

$$\mathbf{X} \in \mathbb{R}^{W \times H}, \{\mathbf{Y}_r \in \{0, 1\}^{W \times H \times K}\}_{r=1}^R; \quad \mathbf{\Psi} \in [0, 1]^{W \times H \times K} = f(\mathbf{X}) \quad (4-6)$$

1392 where the segmentation mask function maps the input to output as:

$$f : \mathbb{R}^{W \times H} \rightarrow [0, 1]^{W \times H \times K} \quad (4-7)$$

1393 The segmentation masks $\mathbf{Y}_r$ satisfy the following condition for being a softmax-like
1394 representation:

$$\mathbf{Y}_r[w, h, :] \mathbf{1}_k^\top = 1; \quad w \in W, h \in H \quad (4-8)$$

1395 4.1.3 Reliability Maps and Truncated GCE

1396 The key innovation in our approach is the introduction of reliability maps Λ_r for
1397 each annotator:

$$\left\{ \Lambda_r(\mathbf{X}; \theta) \in [0, 1]^{W \times H} \right\}_{r=1}^R \quad (4-9)$$

1398 These reliability maps serve as a sophisticated mechanism to estimate the
1399 confidence of each annotator at every spatial location (w, h) in the image. By
1400 learning these maps jointly with the segmentation model, the network gains the
1401 ability to adapt to the varying levels of expertise across different regions of the
1402 image. This approach allows for dynamic weighting of each annotator's

contribution based on their estimated reliability in specific areas, effectively handling cases where annotators might demonstrate different levels of expertise in different parts of the image.

The proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE_{SS}) combines the robustness of GCE with the flexibility of reliability maps:

$$TGCE_{SS}(\mathbf{Y}_r, f(\mathbf{X}; \theta) | \Lambda_r(\mathbf{X}; \theta)) = \mathbb{E}_r \left\{ \mathbb{E}_{w,h} \left\{ \Lambda_r(\mathbf{X}; \theta) \circ \mathbb{E}_k \left\{ \mathbf{Y}_r \circ \left(\frac{\mathbf{1}_{W \times H \times K} - f(\mathbf{X}; \theta)^{\circ q}}{q} \right); k \in K \right\} + \right. \right. \\ \left. \left. (\mathbf{1}_{W \times H} - \Lambda_r(\mathbf{X}; \theta)) \circ \left(\frac{\mathbf{1}_{W \times H} - (\frac{1}{K} \mathbf{1}_{W \times H})^{\circ q}}{q} \right); w \in W, h \in H \right\}; r \in R \right\} \quad (4-10)$$

where $q \in (0, 1)$ controls the MAE or CE level in the same way as in the GCE loss function. The loss function consists of two main components:

- The first term weighted by Λ_r represents the GCE loss for regions where the annotator is considered reliable
- The second term weighted by $(1 - \Lambda_r)$ provides a uniform prior for regions where the annotator is considered unreliable

For a batch containing N samples, the total loss is computed as:

$$\mathcal{L}(\mathbf{Y}_r[n], f(\mathbf{X}[n]; \theta) | \Lambda_r(\mathbf{X}[n]; \theta)) = \frac{1}{N} \sum_n^N TGCE_{SS}(\mathbf{Y}_r[n], f(\mathbf{X}[n]; \theta) | \Lambda_r(\mathbf{X}[n]; \theta)) \quad (4-11)$$

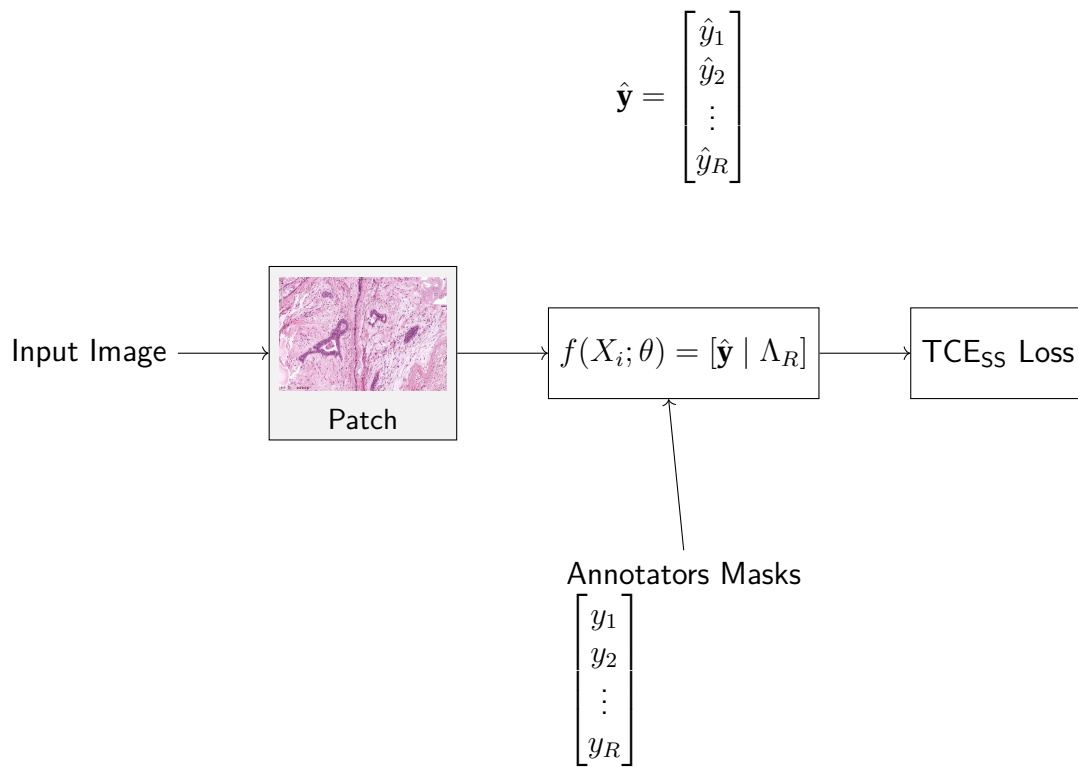


Figure 4-1 Working mechanism of the proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE_{SS}) loss function. The loss combines reliability maps Λ_r with the model's predictions $f(\mathbf{X}; \theta)$ to compute a weighted loss that accounts for annotator reliability across different image regions.

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CHAPTER

FIVE

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CHAINED DEEP LEARNING FOR IMAGE SEGMENTATION

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5.1 Introduction

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As mentioned in Chapter 1, U-shaped CNNs have been proven a good solution for segmentation tasks in medical images, due to their ability to capture both global and local information with relatively low datasets.

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5.2 Using U-Net as a Building Block

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The proposed model architecture combines the strengths of U-Net with a ResNet-34 backbone, specifically designed to work with the $TGCE_{SS}$ loss function by providing a multi-output network which focuses on two tasks: **segmentation** and **reliability estimation**. The architecture is illustrated in Figure 5-2.

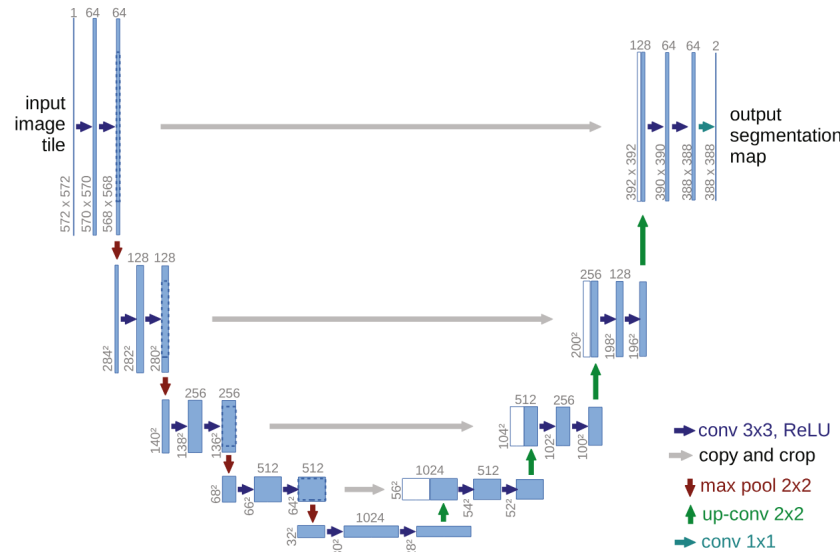


Figure 5-1 Original U-NET architecture.

5.2.1 Backbone Architecture

The model employs a pre-trained ResNet-34 as its encoder backbone, leveraging its deep residual learning framework for efficient feature extraction. The choice of ResNet-34 provides several key advantages: efficient feature extraction through residual connections, pre-trained weights that capture rich visual representations, and stable gradient flow during training. The ResNet-34 backbone is modified to serve as the encoder in our UNET architecture by removing the final fully connected layer and utilizing the feature maps from different stages of the network for skip connections.

5.2.2 U-Net Architecture

The U-Net architecture follows a traditional encoder-decoder structure with skip connections, where the encoder path implements the ResNet-34 structure. The decoder path employs transposed convolutions for upsampling, creating a

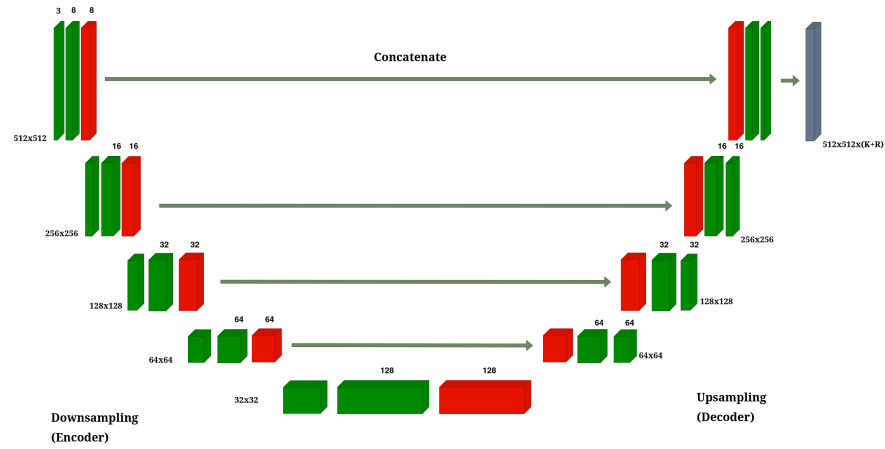


Figure 5-2 Overview of the proposed model architecture.

symmetrical architecture that effectively captures both high-level and low-level features. The architecture incorporates four downsampling stages in the encoder, corresponding to the ResNet-34 blocks, and four upsampling stages in the decoder. These stages are connected through skip connections that bridge corresponding encoder and decoder stages, allowing the network to preserve fine-grained details. Each convolution operation is followed by batch normalization and ReLU activation to ensure stable training and effective feature learning.

This U-Net architecture is expected to be coupled with a reliability branch which leads to a multi-output network. Three different approaches are considered for this branch: **scalar**, **features**, and **pixel**.

5.2.3 Scalar Reliability Estimation

The scalar reliability estimation branch is the simplest one, in which the reliability branch is a parallel branch to the segmentation branch, which leads to a multi-output network. The final output consists of R reliability maps Λ_r , one for each annotator, with values constrained to the $[0, 1]$ range through a sigmoid activation function. This design allows the network to learn and adapt to the varying reliability of different annotators across different regions of the image.

1460 The scalar reliability branch is implemented by taking the bottleneck features from
1461 the encoder and applying a global average pooling operation followed by a dense
1462 layer with sigmoid activation. This results in a single reliability value per annotator
1463 for each input image, making it computationally efficient but less spatially aware of
1464 reliability variations within the image.

1465 5.2.4 Feature-based Reliability Estimation

1466 The feature-based reliability estimation approach provides a more detailed spatial
1467 understanding of annotator reliability by operating on the bottleneck features of
1468 the encoder. This approach maintains the spatial information from the encoder's
1469 bottleneck while being more computationally efficient than pixel-wise reliability
1470 estimation.

1471 The feature-based reliability branch is implemented by applying two convolutional
1472 layers with batch normalization to the bottleneck features:

- 1473 • A first convolutional layer with 32 filters to process the bottleneck features
- 1474 • A second convolutional layer that produces R feature maps, one for each
1475 annotator

1476 This approach provides a coarse-grained spatial reliability map that captures
1477 regional variations in annotator reliability while maintaining computational
1478 efficiency. The resulting reliability maps have the same spatial dimensions as the
1479 bottleneck features, providing a middle ground between scalar and pixel-wise
1480 reliability estimation.

1481 5.2.5 Pixel-wise Reliability Estimation

1482 The pixel-wise reliability estimation approach provides the most detailed spatial
 1483 understanding of annotator reliability by operating on the full-resolution features
 1484 from the decoder. This approach allows for fine-grained reliability estimation at the
 1485 pixel level, capturing local variations in annotator reliability.

1486 The pixel-wise reliability branch is implemented by applying three convolutional
 1487 layers with batch normalization to the decoder’s output features:

- 1488 • Two initial convolutional layers with 32 filters each to process the features
- 1489 • A final convolutional layer that produces R reliability maps, one for each
 1490 annotator

1491 This approach provides the most detailed spatial reliability information, allowing
 1492 the model to capture pixel-level variations in annotator reliability. The resulting
 1493 reliability maps have the same spatial dimensions as the input image, providing the
 1494 finest granularity of reliability estimation among the three approaches.

1495 5.2.6 Integration with TGCE_{SS} Loss

1496 The model produces two distinct outputs: segmentation masks $\hat{\mathbf{Y}} = f(\mathbf{X}; \theta)$ and
 1497 reliability maps $\{\Lambda_r(\mathbf{X}; \theta)\}_{r=1}^R$. These outputs work in tandem with the TGCE_{SS}
 1498 loss function described in Section 4.1. The loss function simultaneously guides the
 1499 learning of both the segmentation masks and reliability maps, ensuring that the
 1500 model learns to balance the contributions of different annotators based on their
 1501 estimated reliability.

1502 The TGCE_{SS} loss function is implemented differently for each reliability approach,
 1503 with specific adaptations to handle the different spatial dimensions of the reliability
 1504 maps. All implementations share common hyperparameters:

- 1505 • $q = 0.1$: Controls the truncation of the loss function
- 1506 • $\epsilon = 10^{-8}$: Small constant to prevent numerical instability
- 1507 • $\alpha_{reg} = 0.1$: Weight for the regularization term
- 1508 • $\alpha_{entropy} = 0.1$: Weight for the entropy regularization
- 1509 • $\alpha_{sum} = 0.1$: Weight for the sum-to-one constraint

1510 For the scalar reliability approach, the loss function operates on global reliability
 1511 values per annotator. The reliability maps are expanded to match the spatial
 1512 dimensions of the predictions through tiling operations. This approach is
 1513 computationally efficient but provides the same reliability value across all spatial
 1514 locations.

1515 The feature-based reliability approach requires additional processing to handle the
 1516 spatial dimensions of the bottleneck features. The reliability maps are resized
 1517 using bilinear interpolation to match the spatial dimensions of the predictions.
 1518 This approach provides coarse-grained spatial reliability information while
 1519 maintaining computational efficiency.

1520 The pixel-wise reliability approach operates directly on full-resolution reliability
 1521 maps, requiring no spatial transformations. This approach provides the most
 1522 detailed spatial reliability information but requires more computational resources.

1523 All three implementations include three regularization terms to ensure proper
 1524 learning of the reliability maps:

- 1525 • **Center Regularization:** $\alpha_{reg} \cdot \mathbb{E}[(\Lambda_r - 0.5)^2]$
- 1526 • **Entropy Regularization:** $-\alpha_{entropy} \cdot \mathbb{E}[\Lambda_r \log(1 + \Lambda_r) + (1 - \Lambda_r) \log(1 + (1 - \Lambda_r))]$
- 1527 • **Sum-to-One Regularization:** $\alpha_{sum} \cdot \mathbb{E}[(\sum_r \Lambda_r - 1)^2]$

1528 These regularization terms help prevent the model from assigning extreme
 1529 reliability values and ensure a balanced distribution of reliability across
 1530 annotators. The loss function also includes a noise tolerance parameter ($\eta = 0.1$)
 1531 to handle cases where the model's predictions differ from the annotators' labels.

1532 The final loss function combines the main $TGCE_{SS}$ terms with the regularization
 1533 terms:

$$\mathcal{L} = \mathbb{E}[\text{term}_1 + \text{term}_2] + \alpha_{reg} + \alpha_{entropy} + \alpha_{sum} \quad (5-1)$$

1534 where term_1 and term_2 are the main $TGCE_{SS}$ terms that handle the interaction
 1535 between predictions and reliability maps.

1536 5.2.7 Training Process

1537 The training process begins with the initialization of the ResNet-34 backbone
 1538 using pre-trained weights, providing a strong foundation for feature extraction.
 1539 The model's hyperparameters are optimized using Bayesian optimization through
 1540 Keras Tuner, which searches for optimal values of:

- 1541 • Learning rate: Sampled logarithmically between 10^{-5} and 10^{-2}
- 1542 • $TGCE_{SS}$ parameters:
 - 1543 - q : Controls truncation of the loss function (0.1 to 0.9)
 - 1544 - Noise tolerance: Controls model's robustness to label noise (0.1 to 0.9)
 - 1545 - Regularization weights: Control the balance between different loss
 - 1546 terms (0.01 to 0.5)

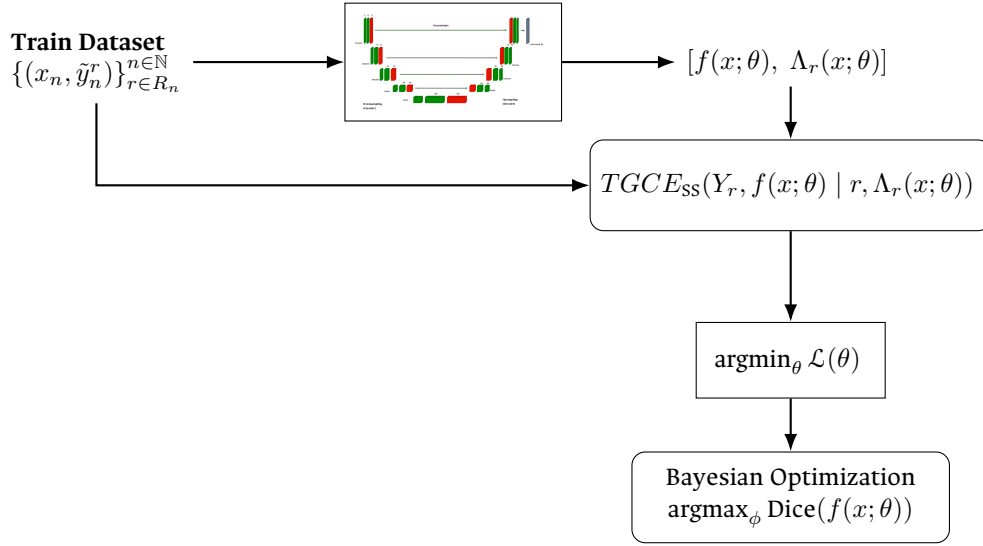


Figure 5-3 Training process of the proposed model overview. The model first undergoes hyperparameter optimization using Bayesian optimization, followed by end-to-end training with the optimal parameters. Estimated segmentation and reliability maps are computed for each input image, and the loss function is computed for the entire batch. Network parameters are tuned based on the loss function as objective (with minimization target) whereas the hyperparameters ϕ tuning are based on the Dice coefficient of the validation data set (with maximization target).

1547 The hyperparameter search is performed using a validation set, with the objective
 1548 of maximizing the Dice coefficient on the validation data. After finding the optimal
 1549 hyperparameters, the model is trained end-to-end using the Adam optimizer with
 1550 the best learning rate found. The $TGCE_{SS}$ loss function plays a crucial role in
 1551 updating both the segmentation and reliability branches, ensuring that the model
 1552 learns to effectively handle multiple annotators' inputs while accounting for their
 1553 varying reliability.

1554 The model's architecture is specifically designed to address the challenges of
 1555 multi-annotator segmentation. Through the ResNet-34 backbone, it learns robust
 1556 segmentation features that capture high-level patterns in the data. The UNET's
 1557 skip connections enable the preservation of fine-grained details, while the parallel
 1558 reliability branch allows the model to adapt to annotator-specific characteristics.
 1559 This comprehensive design enables the model to effectively handle multiple

1560 annotators' inputs while maintaining high segmentation accuracy and reliability
1561 estimation.

1562 5.3 Experimental Setup

1563 5.3.1 Noisy Annotations Generation

1564 A challenge arises for the creation of emulated noisy annotations datasets, since it
1565 is expected for images annotations to have some degree of expertise variability,
1566 similar to what is expected in real crowdsourced datasets. Simply introducing
1567 random noise into the annotated masks does not work, since the original
1568 morphological structures from the expected ground truth are far from being
1569 preserved. Instead, a “noisy” labeler is expected to produce an annotation which
1570 has at least some degree of morphological consistency on itself, even if it shows
1571 discrepancies when compared with some metric (like DICE score) against the
1572 ground truth annotation.

1573 This has been proven experimentally when introducing random noise into any
1574 popular segmentation dataset, in which the resulting mask is just a non coherent
1575 map of noise across the image, as shown in Figure 5-4.

1576 For this reason, a more sophisticated approach was needed to create datasets with
1577 emulated noisy annotations. The approach used here was to use a pre-trained
1578 U-Net model to produce a good enough ¹ segmentation of the image, which was
1579 then used to produce a “noisy” annotation by introducing random noise into the
1580 last encoder layers of the model, thus preserving the morphological consistency of
1581 the original annotation. This strategy works since slight modifications introduced

¹More precisely, a U-Net architecture with Resnet-34 backbone model with a simple categorical cross-entropy loss which reached 0.8426 Dice Coefficient after 44 epochs.

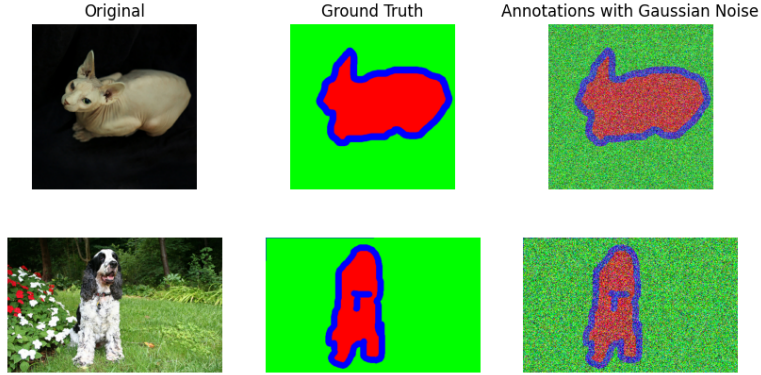


Figure 5-4 Example of a noisy mask generated by naively introducing random noise into a ground truth mask. Morphological consistency is lost.

1582 into the encoder layers weights, somehow resemble conceptual disturbances with
 1583 respect to the respect to the original ground truth label, which kind of emulates
 1584 the cultural behavior of having a different interpretation of the image, either for
 1585 having a different level of expertise or for having a different point of view or
 1586 school of thought. The first encoding layers were not modified, since it is expected
 1587 human labelers would agree on the most fundamental structures (analog to
 1588 extracted features from initial convolutional layers) in a similar way, thus
 1589 preserving the morphological consistency of the original annotation.

1590 In this way, the level of “disturbance” with respect to the ground truth for an
 1591 emulated annotator can be controlled by the level of noise introduced into the
 1592 weights of the last encoder layers, thus:

$$\mathbf{W}_{noisy} = \mathbf{W}_{original} + \epsilon_d \quad (5-2)$$

1593 where $\mathbf{W}_{original}$ represents the original weights of the last encoder layers, ϵ_d follows
 1594 a Gaussian distribution with zero mean and variance σ^2 . The variance σ^2 controls
 1595 the level of noise introduced, and thus the degree of disturbance in the resulting
 1596 segmentation masks.

With the application of the encoder layer weight perturbation technique, the resulting noisy masks are shown in Figure 5-5. It can be seen that the morphological consistency is preserved, even though the resulting masks are far from the ground truth annotations, which goes perfectly well for testing the robustness of the models against noisy annotations in crowdsourcing-like scenarios.

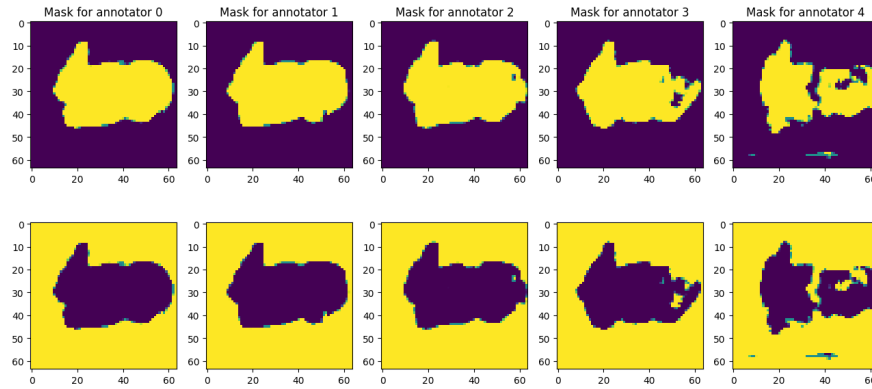


Figure 5-5 Noisy mask generated by enhancing the disturbances in the encoder layers weights for the Oxford-IIIT Pet Dataset. Morphological consistency is preserved. From left to right, SNR levels of noise in the encoder layer are 10, 5, 2, 0, -5 dB. Not adding noisy to the encoder layer would reach to an annotator behavior as good as the originally trained base model.

5.3.2 Missing Labels Emulation

It has also been determined that most real world crowdsourced datasets suffer from missing labels, since not all annotators are able to label all of the input samples. To emulate this behaviour, a missing labels dataset was created by randomly skipping labels from the annotators labels based on a labeling rate $r_l \in [0, 1]$. The labeling rate r_l controls the proportion of labels that are skipped from the annotators labels. The lower the labeling rate, the more labels are skipped from the annotators labels. The implementation of this missing labels emulation together with the noisy annotators generation can be flexibilized through an algorithm, which enables it to be used in

1612 experiments with variable amount of labelers, noise levels, and labeling rates. An
 1613 overview of the algorithm is shown in Algorithm 5-1. A sample of the generated
 1614 noisy and missing labels dataset is shown in Figure 5-6.

Algorithm 5-1 Multi-Annotator Dataset Generation with Noisy and Missing Labels

Require: Base segmentation model M , SNR levels $\{s_1, \dots, s_n\}$, labeling rate $r \in [0, 1]$

Ensure: Dataset with multiple noisy annotations and missing labels

```

1: Step 1: Generate Noisy Annotators
2: for each SNR level  $s_i$  do
3:   Create disturbed model  $M_i$  by adding noise to  $M$  with SNR  $s_i$ 
4: end for
5: Step 2: Initialize Labeler Assignment System
6: Create assignment matrix  $A \in \{0, 1\}^{N \times n}$  where  $N$  is number of samples
7: for each annotator  $j \in \{1, \dots, n\}$  do
8:   for each sample  $i \in \{1, \dots, N\}$  do
9:      $A_{i,j} \leftarrow \begin{cases} 1 & \text{with probability } r \\ 0 & \text{with probability } 1 - r \end{cases}$ 
10:   end for
11: end for
12: Step 3: Generate Multi-Annotator Dataset
13: for each image  $x$  in dataset do
14:   Resize  $x$  to target dimensions
15:   for each annotator model  $M_i$  do
16:     Generate annotation  $y_i = M_i(x)$ 
17:     Apply labeler assignment mask:  $y_i \leftarrow y_i \cdot A_{i,j}$ 
18:   end for
19:   Store  $(x, \{y_1, \dots, y_n\})$  in dataset
20: end for
   return Multi-annotator dataset with noisy and missing labels

```

Method	Dice	Jaccard
TGCE scalar	0.8564	0.7691
TGCE pixel	0.8334	0.7392
STAPLE	0.8639	0.7795
MV	0.8214	0.7351

Table 5-1 Performance comparison of methods using Dice and Jaccard metrics for the Oxford-IIIT Pet Dataset with no labelers aggregation. STAPLE outperforms all other methods, followed by TGCE_{SS} scalar and pixel. MV performs the worst.

5.4 Results

5.4.1 Oxford-IIIT Pet Dataset

The experimental results on the Oxford-IIIT Pet Dataset demonstrate interesting insights into the performance of different approaches. While STAPLE achieves the highest Dice and Jaccard scores (0.8639 and 0.7795 respectively), our proposed TGCE_{SS} approaches show competitive performance, with the scalar variant achieving a Dice score of 0.8564 and Jaccard of 0.7691. The pixel-wise variant, while slightly lower in performance metrics, provides more detailed spatial reliability information.

It's important to note that these results, when compared directly with STAPLE and Majority Voting (MV) without labeler aggregation, show that STAPLE maintains its advantage in terms of pure segmentation accuracy. However, this comparison alone doesn't fully capture the value of our approach. The key strength of our method lies in its ability to simultaneously learn both segmentation and reliability estimation, providing valuable insights into annotator behavior and confidence levels that STAPLE and MV cannot offer.

The reliability estimation capabilities of our model are particularly noteworthy, as demonstrated in Figures 5-7 and 5-8. The model successfully captures the varying reliability levels of different annotators, with higher reliability assigned to annotators with lower noise levels (10 dB) and lower reliability to those with higher noise levels (-20 dB). This ability to quantify and visualize annotator

1636 reliability is a significant advantage over traditional approaches like STAPLE and
1637 MV, which focus solely on segmentation accuracy without providing insights into
1638 annotator behavior.

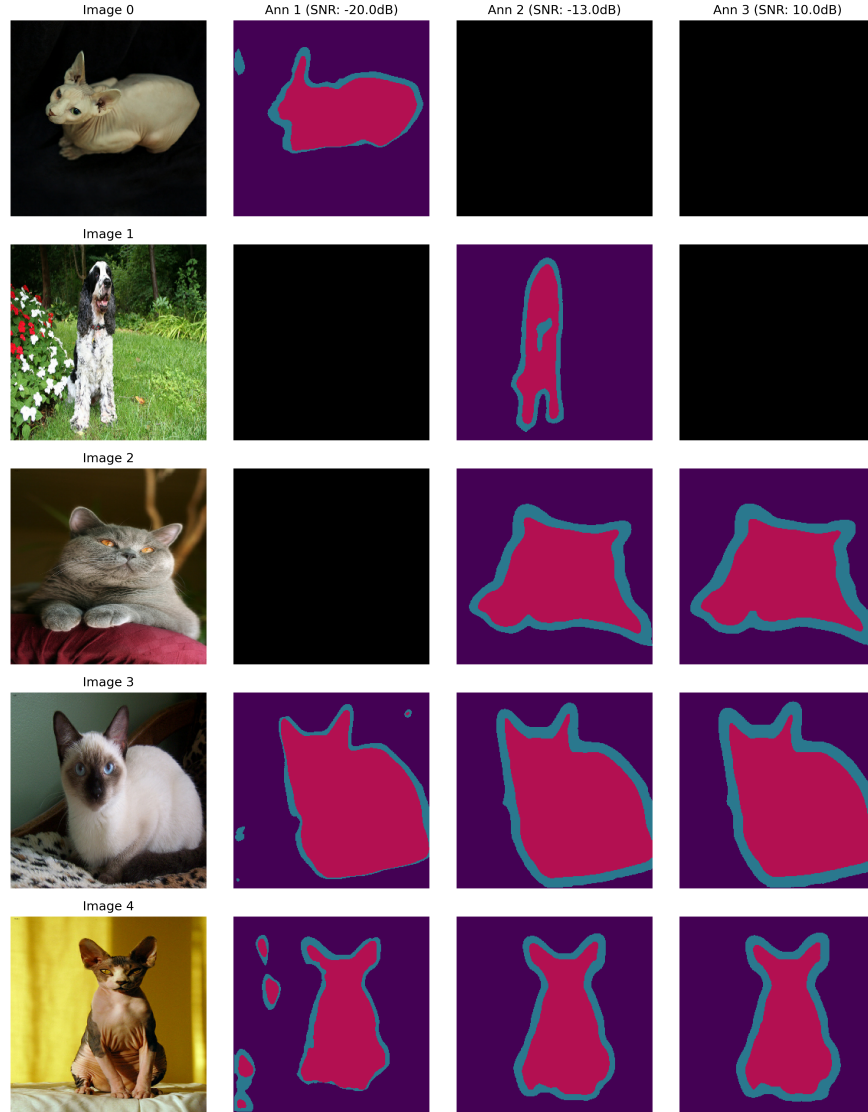


Figure 5-6 Example of missing labels and noisy annotations emulation for the Oxford-IIIT Pet Dataset. The same label rate ($r_l = 0.7$) is preserved for all labelers, which means in this case that every labeler is expected to label 70% of the samples, and skip the remaining 30%.

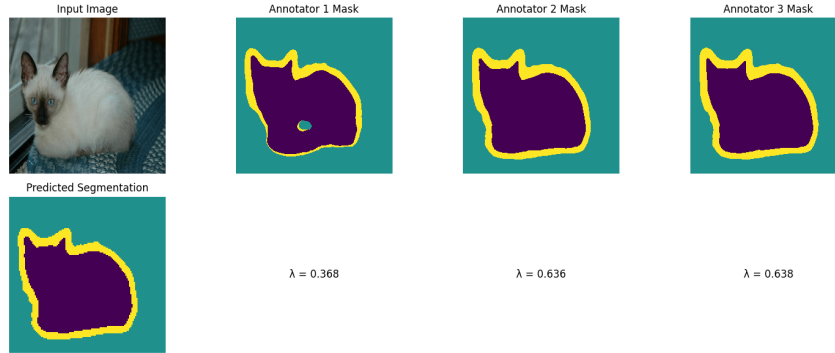


Figure 5-7 Example evaluation of the proposed model in its last trained epoch. An accurate segmentation output is delivered as well as scalar reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0011, $q = 0.4$, noise tolerance = 0.2, $\alpha_{reg} = 0.26$, $\alpha_{entropy} = 0.03$, $\alpha_{sum} = 0.27$.

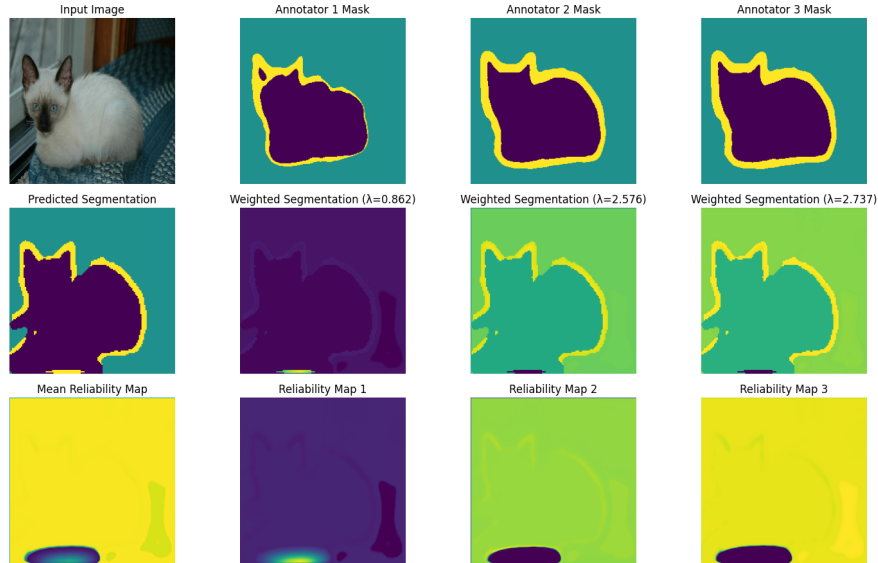


Figure 5-8 Example evaluation of the proposed model in some given epoch (5/50) of training, which demonstrates a quick convergence of the model for correctly estimating the reliability of the labelers masks. An accurate segmentation output is delivered as well as pixel-wise reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0027, $q = 0.9$, noise tolerance = 0.2, $\alpha_{reg} = 0.38$, $\alpha_{entropy} = 0.02$, $\alpha_{sum} = 0.42$.

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CHAPTER

SIX

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FINAL REMARKS

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6.1 Conclusions

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Throughout this work, several techniques have been explored to develop a model that can face **ISS** of real world histopathology images in crowdsourcing scenarios. Different approaches were studied and evaluated to reach an optimal solution which could model the annotators' performance from the input space. The following remarks are based on the results obtained from the experiments conducted and reviewed literature:

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- Chained Gaussian Processes approaches, while robust for classification and regression tasks, proved computationally intensive for image segmentation. This limitation highlighted the need for more efficient approaches that could handle the complexity of medical image segmentation while modeling annotator performance.

- The development of a novel loss function was crucial for the success of this work, as existing loss functions were insufficient to model the annotators' performance across the input space. The proposed loss function successfully enabled the model to adaptively infer the best possible segmentation without requiring separate inputs about labeler performance.
- The introduction of a tensor map to codify labeler reliability proved to be an effective approach, allowing the model to implicitly weigh the labelers based on their performance across the mask and classes space, thus handling the labeler performance inconsistency effectively.
- The use of U-Shaped Deep Learning models as a building block for the proposed solution was validated, as it effectively captured spatial relationships between pixels and annotations while handling the variability in image quality.
- The combination of Chained Deep Learning approaches with the proposed loss function and tensor map representation resulted in a comprehensive solution being, to the best of our knowledge, the first approach on offering an image segmentation solution for multiple labelers in medical image with interpretability in the reliability of the labelers, successfully addressing both labeler inconsistency and image quality variability.

6.2 Future work

As tools and frameworks for bayesian approaches like Chained Gaussian Processes¹ evolve, it is expected to see a decrease in computational requirements and hence, making mixed models more appealing for the task of ISS of histopathology images. In that sense, the use of CCGPMA as a building block for the proposed solution could

¹Including GPFlow, and GPFlux.

be a good approach, as it is a powerful model that can take advantage of the inter-dependencies between the annotators, and the powerfulness of CNNs to extract features from the images.

The noisy labelers emulation technique developed in this work presents an opportunity to test the model's robustness in various medical image segmentation challenges. This technique could be adapted to simulate different types of expert variability in other medical imaging domains:

- **Multi-modal Medical Imaging:** The technique could be extended to simulate expert disagreements in multi-modal scenarios, where different imaging modalities (e.g., MRI, CT, PET) might lead to varying interpretations of the same anatomical structure.
- **Temporal Consistency:** For longitudinal studies, the emulation could be modified to simulate how expert annotations might vary over time, helping to test the model's ability to maintain consistency in segmentation across different time points.
- **Cross-institutional Variability:** The noise injection technique could be calibrated to simulate the variations in annotation patterns that arise from different institutional protocols or training backgrounds.
- **Pathology-specific Challenges:** The emulation could be tailored to simulate expert disagreements in specific pathological conditions, where the morphological features might be more ambiguous or subject to interpretation.

These extensions would help validate the model's performance across a broader range of medical imaging scenarios and ensure its robustness in real-world clinical applications.

The pixel-wise reliability mapping approach shows particular promise for future development. While the current implementation already demonstrates good performance, several enhancements could lead to significant improvements:

1708 • **Advanced Data Augmentation:** Implementing a comprehensive data
1709 augmentation strategy could significantly improve the model's convergence
1710 and performance. This could include:

- 1711 – Geometric transformations (rotations, flips, elastic deformations)
- 1712 – Intensity variations to simulate different staining conditions
- 1713 – Multi-scale training to handle varying tissue sizes
- 1714 – Adversarial training to improve robustness

1715 These augmentations would help the model learn more robust features and
1716 better generalize to unseen data.

1717 • **Attention Mechanisms:** Integrating attention mechanisms into the pixel-wise
1718 reliability branch could help the model focus on the most relevant regions for
1719 reliability estimation, potentially improving both accuracy and computational
1720 efficiency.

1721 • **Multi-scale Reliability Estimation:** Implementing a hierarchical approach to
1722 reliability estimation could capture both fine-grained local variations and
1723 broader regional patterns in annotator reliability, this would imply a
1724 network with yet more branches and outputs.

1725 The combination of these enhancements with the existing pixel-wise reliability
1726 mapping approach could potentially outperform all other techniques, as it would
1727 provide the most detailed and accurate representation of annotator reliability
1728 while maintaining computational efficiency. The use of data augmentation, in
1729 particular, could help the model converge faster and achieve better generalization,
1730 as it would expose the model to a wider variety of possible variations in both the
1731 input images and the corresponding annotations.

6.3 Academic Discussion

To facilitate the adoption and further development of the proposed approach, several resources have been made publicly available. A comprehensive GitHub repository has been created that provides easy-to-use APIs for implementing the proposed solution. The repository includes pre-built dataset loaders for TensorFlow/Keras frameworks, implementation of the $TGCE_{SS}$ loss function in all its variants (scalar, features, and pixel-wise), and a neural network architecture builder with experimental examples. A key feature of this repository is its automatic dataset fetching from AWS S3, which eliminates the need for manual downloads or external dependencies. This repository aims to lower the entry barrier for researchers and practitioners interested in implementing the proposed approach.

To further simplify the adoption of the proposed solution, a PyPI package has been published with detailed documentation available at <https://seg-tgce.readthedocs.io/en/latest/>. The documentation provides comprehensive installation instructions, API reference, usage examples, and best practices for implementation. This package makes it straightforward for the community to integrate the proposed approach into their existing workflows, with a focus on ease of use and clear documentation of all available features and options.

An academic publication is currently being prepared for submission to a Q1 journal. This publication will provide detailed theoretical foundations of the proposed approach, comprehensive experimental results and analysis, comparison with state-of-the-art methods, and discussion of limitations and future directions. The publication will serve as the primary reference for the theoretical and practical aspects of the proposed solution, ensuring that the work is properly documented and validated through the peer-review process.

These resources collectively contribute to the reproducibility and accessibility of the research, enabling the broader scientific community to build upon and extend

1761 the proposed approach. The combination of open-source code, well-documented
1762 APIs, and peer-reviewed publication ensures that the work can be properly
1763 evaluated, validated, and utilized by other researchers in the field. This
1764 comprehensive approach to sharing research outputs aligns with current best
1765 practices in scientific computing and machine learning research, promoting both
1766 transparency and practical utility of the proposed solution.

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